WILL

Relational Orbital Mechanics (R.O.M.)

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Abstract

This supplement to the first part of the WILL trilogy presents **Relational Orbital Mechanics (R.O.M.)** – a closed algebraic system that classifies allowed bound states of relational configuration without invoking a background spacetime, a metric tensor, or a mass and gravitational constant.

From the dimensionless projections β (kinematic) and κ (potential) on the relational carriers S^1 and S^2, and from the closure theorem $\kappa^2 = 2\beta^2$ (forced by degree-of-freedom indifference), all major observed orbital phenomena are derived algebraically as indefinable consequence of Relational Geometry:

- Kepler's Third Law, the vis-viva relation, conservation of angular momentum, and the derivation of eccentricity emerge as strict Pythagorean invariants.
- Orbital precession is recovered as the accumulation of the relational phase divergence τ_Y^2, normalized by the orbital shape factor $1 - e^2$, yielding the exact GR value for Mercury without differential equations.
- The Schwarzschild radius R_s and all orbital scales are expressed directly in terms of spectroscopic and chronometric observables ($z_\odot$, $\theta_\odot$, T, e), bypassing mass M and Newton's constant G.
- Gravitational deflection of massive particles and light follows from the energy symmetry law with no weak-field expansion.

The results are tested against the full 24-year astrometric and radial-velocity dataset of the S2 star around Sgr A*. Using the same parameters as a standard General Relativity 1PN model, the R.O.M. parametrization – which couples the pericentre advance to the true anomaly rather than to coordinate time – achieves a $\Delta\chi^2 \approx -74$ improvement. All orbital scales are extracted from the dimensionless observables alone.

R.O.M. demonstrates that the operational content of gravitational dynamics is fully captured by the algebra of relational projections. The construct serves as a rigorous, empirical realization of the foundational principles derived in WILL Part I: Relational Geometry. As an Open Research all data and calculation tools are publicly available at willrg.com and https://github.com/AntonRize/WILL.

*This work is archived on Zenodo: ¸

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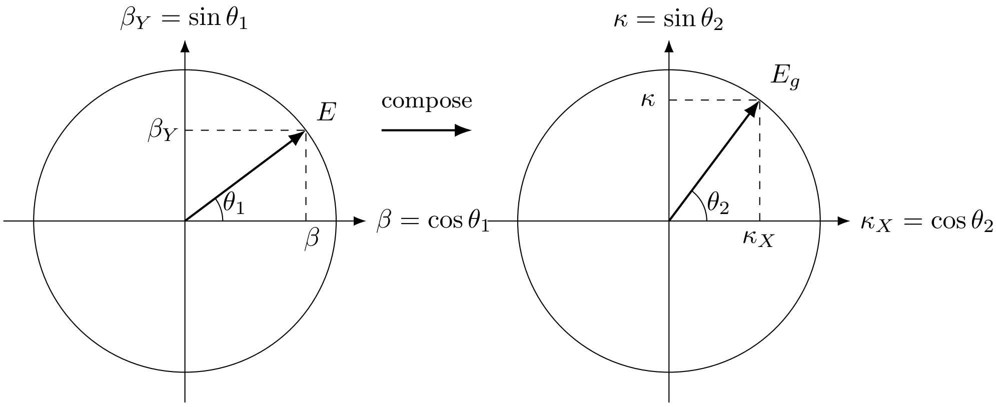
1 Quick Recap

In following sections we will demonstrate that RG's relational projections (κ, β) are ontologically and operationally independent from mass M and gravitational constant G. We will show that R_s as a horizon scale derived directly from the phase relations of light and will show how all observational phenomenon of orbital systems are generated as the unavoidable consequences of Relational Geometry.

But first lets remind our selves key derivations and logical chain established prior:

Logical Chain Summary

Methodological Constraints
$\downarrow$ *(apply to existing physics)*
False Separation Identified
$\downarrow$ *(dissolve)*
SPACETIME $\equiv$ ENERGY
$\downarrow$ *(derive consequences)*
Closure + Conservation + Isotropy
$\downarrow$ *(classify carriers)*
S^1 *(1-DOF kinematic)* + S^2 *(2-DOF potential)*



$E = E_0/\beta_Y$
$p = E_0\,\beta/\beta_Y = E_0\cot(\theta_1)$

$E_g = E_0/\kappa_X$
$p_g = E_0\,\kappa/\kappa_X = E_0\tan(\theta_2)$

$\theta_1 = \arccos(\beta)$, $\quad\theta_2 = \arcsin(\kappa)$, $\kappa^2 = 2\beta^2$	
Algebraic Form	**Trigonometric Form**
$\beta = v/c$	$\beta = \cos(\theta_1)$
$\beta_Y = \sqrt{1-\beta^2}$	$\beta_Y = \sin(\theta_1)$
$\kappa = \sqrt{R_s/r}$	$\kappa = \sin(\theta_2)$
$\beta_Y = \sqrt{1-\beta^2}$	$\beta_Y = \sin(\theta_1)$
$\kappa_X = \sqrt{1-\kappa^2}$	$\kappa_X = \cos(\theta_2)$
$p = E_0/c \cdot \beta/\beta_Y$	$p = E_0/c \cdot \cot(\theta_1)$
$p_g = E_0/c \cdot \kappa/\kappa_X$	$p_g = E_0/c \cdot \tan(\theta_2)$
$\tau = \beta_Y\kappa_X$	$\tau = \sin(\theta_1)\cos(\theta_2)$
$\tau_Y = \sqrt{1-\tau^2}$	$\tau_Y = \sqrt{1-\tau^2}$
$Q = \sqrt{\kappa^2+\beta^2} = \sqrt{3}\beta$	$Q = \sqrt{3}\cos(\theta_1)$

Table 1: Unified representation of relativistic and gravitational effects for closed systems.

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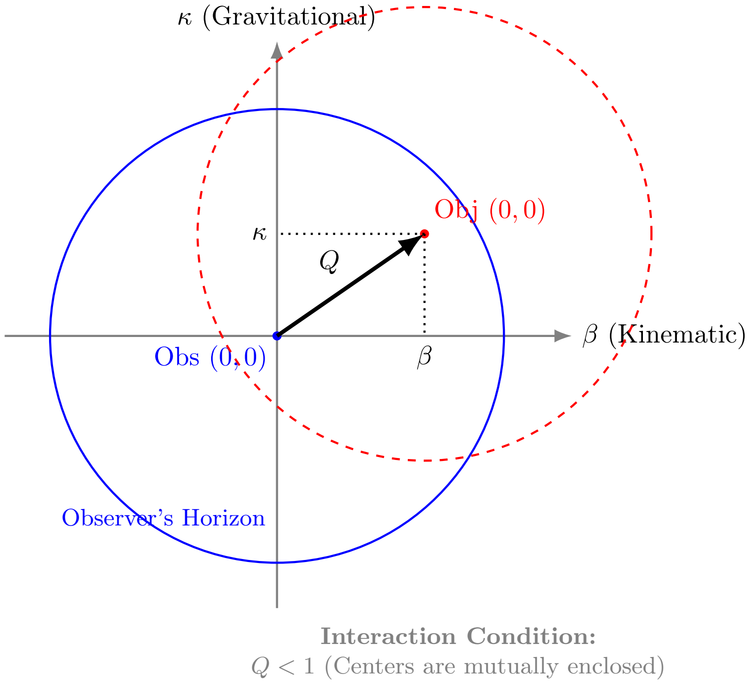
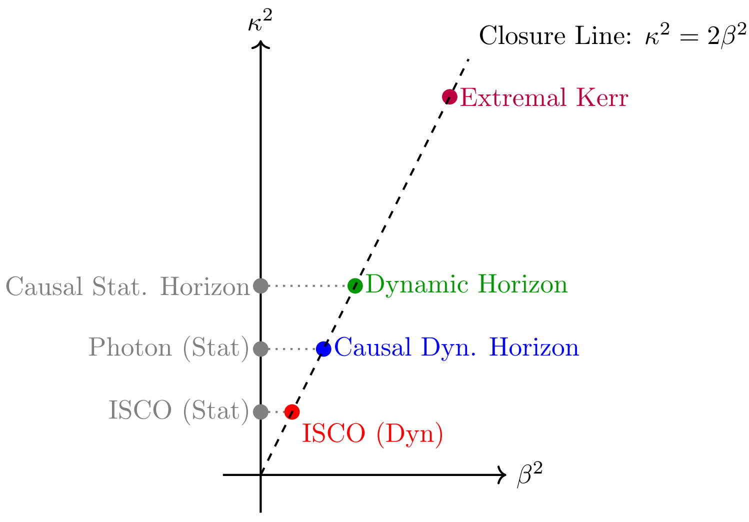


Figure 1: **Relational Self-Centering.** The total shift Q is defined by the orthogonal projections β and κ. Interaction is causal only when the center of the Object lies within the Observer's horizon ($Q < 1$), ensuring mutual coverage.

Phenomenon	**Radius** r	β^2	κ^2	Q^2	**Comment**
ISCO (Static location)	$3R_s$	0	$\frac{1}{3}$	$\frac{1}{3}$	Pure coordinate location, $r = R_s/\kappa^2$
ISCO (Dynamic state)	$3R_s$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	Marginal orbital stability, $Q^2 = Q_Y^2 = 1/2$
Photon sphere (Static)	$\frac{3}{2}R_s$	0	$\frac{2}{3}$	$\frac{2}{3}$	Pure coordinate location
Causal Dynamic horizon	$\frac{3}{2}R_s$	$\frac{1}{3}$	$\frac{2}{3}$	1	Relational horizon under closure $\kappa^2 = 2\beta^2$, $Q = 1$
Causal Static horizon	R_s	0	1	1	Purely gravitational closure limit, $\kappa^2 = 1$
Dynamic horizon	R_s	$\frac{1}{2}$	1	$\frac{3}{2}$	Orbital closure at Schwarzschild radius
Extremal Kerr horizon	$\frac{1}{2}R_s$	1	2	3	Maximal rotation, $\beta = 1$, merged horizons

Table 2: Critical radii and their projectional parameters. Static coordinate locations ($\beta = 0$) are strictly distinguished from their corresponding dynamic orbital states that emerge from the closure law $\kappa^2 = 2\beta^2$.



2 Closed Algebraical System of R.O.M. Equations

Desmos Project

R.O.M.

Remark 2.1. *R.O.M. does not describe how a body moves under forces; it classifies the algebraically allowed relational states of a bound system. It is not equations of motion but algebraically closed system of allowed states within WILL - allowed free will.*

$\boxed{\kappa^2 = 1 - \frac{1}{(1+z_\kappa)^2}}$ z_κ = gravitational redshift

$\boxed{\beta^2 = 1 - \frac{1}{(1+z_\beta)^2}}$ z_β = transverse Doppler shift

Observational Inputs

$Z_{sys} = (1+z_\kappa)(1+z_\beta) = \frac{1}{\tau}$ (product of gravitational red shift and transverse Doppler shift)
$\tau = \kappa_X \cdot \beta_Y = \frac{1}{Z_{sys}}$ (product of projectional phase factors on S^1 and S^2 carriers)
$z_\kappa = \frac{1}{\kappa_X} - 1$ (gravitational redshift)
$z_\beta = \frac{1}{\beta_Y} - 1$ (transverse Doppler shift)
$z_{\kappa o} = \frac{1}{\kappa_{Xo}} - 1$ (redshift at phase O)
$z_{\beta o} = \frac{1}{\beta_{Yo}} - 1$ (transverse Doppler shift at phase O)

Global System Parameters (Fixed for the Orbit)

$\kappa = \sin(\theta_2) = \sqrt{1-(1+z_\kappa)^{-2}} = \sqrt{\frac{R_s}{a}} = \sqrt{\frac{\rho}{\rho_{max}}} = \sqrt{2}(\frac{\pi R_s}{Tc})^{\frac{1}{3}} = \sqrt{\kappa_p^2(1-e)} = \sqrt{2(\kappa_o(O)^2 - \beta_o(O)^2)} = \sqrt{\frac{1}{2}(3 - \sqrt{1+8\tau_o(B_a)}$
$\kappa_R^2(\frac{I_t}{T})^{\frac{2}{3}} = \sqrt{\kappa_o^2 \cdot \eta_o}$ (potential projection at semi-major axis)
$\kappa_X = \cos(\theta_2) = \sqrt{1-\kappa^2} = \frac{1}{(1+z_\kappa)}$ (potential phase factor)
$\kappa_R = \sqrt{\frac{\kappa^2}{\sin(\theta_R)}} = \sqrt{\frac{R_s}{\frac{T}{2\pi} c\beta \sin(\theta_R)}}$ (potential projection at the surface of)
$\beta = \cos(\theta_1) = \sqrt{1-(1+z_\beta)^{-2}} = \frac{\kappa}{\sqrt{2}} = \beta_p\sqrt{\frac{1-e}{1+e}} = \frac{2\pi a}{Tc} = (\frac{\pi R_s}{Tc})^{\frac{1}{3}} = \sqrt{(\kappa_o(O)^2 - \beta_o(O)^2)} = \sqrt{\frac{\kappa_p^2}{2}(1-e)} = \beta_o(O)\frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2\cdot e\cdot\cos(O)}} = \sqrt{\beta_{sys}^2 \cdot \sin(\theta_{sys})}$ (kinetic projection on semi major axis)
$\beta_Y = \sin(\theta_1) = \sqrt{1-\beta^2}$ (kinetic phase factor)
$\beta_{int} = \frac{\beta}{\sqrt{1-e^2}}$ (interior kinetic projection)
$\tau = \kappa_X \beta_Y$ (relational spacetime factor)
$\tau_Y = \sqrt{1-\tau^2} = \sqrt{3\beta^2 - 2\beta^4}$ (relational spacetime phase factor)
$Q = \sqrt{\kappa^2+\beta^2} = \sqrt{\frac{3}{2}}\ \kappa = \sqrt{3}\ \beta$ (total relational shift (magnitude of state difference))
$R_s = \kappa_o(O)^2 r_o(O) = \frac{Tc}{\pi}\beta^3 = \frac{8\pi^2 a^3}{T^2c^2} = \frac{\Delta_{to}(O)}{\zeta(O)}\kappa^2\beta c = \frac{r_1 r_2}{r_2 - r_1}(\beta_1^2 - \beta_2^2) = \frac{a}{2}(3 - \sqrt{1+8\tau_o(B_a)^2}) = \frac{r_o(O)}{2(2a - r_o(O))}(4a - r_o(O) - \sqrt{(4a - r_o(O))^2 - 8a(2a - r_o(O))(1-\tau_o(O)^2)}) = \frac{2GM}{c^2}$ (Schwarzschild radius - system scale)
$a = \frac{R_s}{\kappa^2} = \frac{T\beta c}{2\pi} = \frac{\beta_o c}{\omega} \cdot \frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2e\ cos(O)}} = \frac{\Delta_{to}(O)}{\zeta}\beta c = \frac{Tc\beta^3}{2\pi(\frac{K_i}{\sin(i)})^2(1-e^2)} = \frac{\sqrt{1-e^2}}{2\pi\sin(i)}K_i Tc = \frac{R_{sys}}{\sin(\theta_{sys})}$ (semi-major axis)
$R_{sys} = a \cdot \sin(\theta_{sys})$ (system's central body's physical radius)
$M = \frac{\kappa^2 c^2 a}{2G} = 4\pi\rho a^3$ (mass parameter)
$t = \frac{a}{c}$(temporal scale of the system)
$\beta^2 = (\kappa_o^2 - \beta_o^2) = \frac{\kappa_p^2}{2}(1-e) = (\kappa_o(O)^2 - \frac{\kappa_o(O)^2}{2\delta_o(O)}) = \frac{R_s}{2a} = (\frac{R_s}{T\cdot c}\pi)^{\frac{2}{3}}$ (energy invariant - binding energy)
$\Delta\phi = \frac{2\pi\cdot\tau_Y^2}{1-e^2} = \frac{2\pi(3\beta^2 - 2\beta^4)}{1-e^2} = \frac{3\pi R_s}{a(1-e^2)} - \frac{\pi R_s^2}{a^2(1-e^2)}$ (precession of perihelion per orbit)
$\Omega_\phi = \frac{\tau_Y^2}{1-e^2} = \frac{3\beta^2 - 2\beta^4}{1-e^2}$ (precession of perihelion per phase radian)
$\Omega = 1 - \Omega_\phi = \frac{O}{o}$ (constant ratio between idealised phase o and true precessing phase O)
$h_W = r_o(O)\beta_T(O)c = r_o^2 \cdot \omega_o = a \cdot \beta c \cdot e_Y = R_s c\frac{\sqrt{1-e^2}}{2\beta}$ (invariant angular momentum)
$\omega = \frac{\beta c}{a} = \frac{c}{R_s}2\beta^3$ (angular frequency)
$T = \frac{2\pi}{\omega} = \frac{2\pi a}{\beta c} = \frac{\pi R_s}{c\beta^3} = \frac{\Delta_{to}}{\zeta_o}2\pi$ (orbital period)

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$\Delta\varphi = 2\arcsin(\frac{\kappa_p^2(1+\beta_p^2)}{2\beta_p^2-\kappa_p^2(1+\beta_p^2)})$ (gravitational deflection)
$\Delta\gamma = 2\arcsin(\frac{\kappa_p^2}{\kappa_{Xp}^2})$ (light deflection)

Eccentricity Relations

$e = \frac{1}{\delta_p} - 1 = 1 - \frac{2\beta_a^2}{\kappa_a^2} = \frac{2\beta_p^2}{\kappa_p^2} - 1 = 1 - \eta_o(0) = \eta_o(\pi) - 1$ (eccentricity derived from closure)
$e_Y = \sqrt{1-e^2}$ (eccentricity's orthogonal value)
$e_X = \frac{1+e}{1-e} = \frac{r_a}{r_p} = \frac{\delta_a}{\delta_p} = \frac{\beta_p}{\beta_a} = \frac{\kappa_p^2}{\kappa_a^2} = \frac{\kappa_a^2\beta_p^2}{\kappa_p^2\beta_a^2}$ (shape factor, still looks like magic to me)

Time-phase

$\omega_o(O) = \frac{\beta \cdot c}{a} \cdot \frac{(1+e\cdot cos(O))^2}{(1-e^2)^{\frac{3}{2}}}$ (angular frequency at phase o)
$\Delta_{to}(O) = \int_0^O \frac{1}{\omega_\theta(\theta)}\,d\theta = \frac{a}{\beta c}\zeta = \frac{T}{2\pi}\zeta = \frac{R_s\zeta}{2\beta^3 c}$ (time duration of given phase interval)
$\zeta(O) = (1-e^2)^{\frac{3}{2}} \cdot (\int_0^O (1+e\cdot\cos(\theta))^{-2} d\theta) = \frac{1}{\sqrt{1-e^2}} \int_0^O (\frac{t_o(\theta)}{t})^2 d\theta = \frac{2\pi\Delta_t}{T}$ (Temporal Phase interval)
$\zeta(<\pi) = 2\arctan(\frac{\sin(\frac{O}{2})}{e_X^{\frac{1}{2}}\cos(\frac{O}{2})}) - \frac{e\cdot e_Y \sin(O)}{1+e\cos(O)}$ (Temporal Phase interval on ascending orbit from 0 to π)
$\zeta(>\pi) = 2\arctan(\frac{\sin(\frac{O}{2})}{e_X^{\frac{1}{2}}\cos(\frac{O}{2})}) - \frac{e\cdot e_Y \sin(O)}{1+e\cos(O)} + 2\pi$ (Temporal Phase interval on descending orbit from π to 0)

Perihelion Relations

$r_p = a(1-e) = \frac{R_s}{\kappa_p^2} = \sqrt{((1-e)\cdot a\cdot\cos(\Omega_O o))^2 + ((1-e)\cdot a\cdot\sin(\Omega_O o))^2}$ (radius at perihelion)
$\kappa_p = \kappa\sqrt{\frac{1}{1-e}} = \sqrt{\frac{2\beta^2}{1-e}} = Q_p\sqrt{\frac{2}{3+e}}$ (potential projection at perihelion)
$\kappa_{Xp} = \sqrt{1-\kappa_p^2}$ (potential phase projection at perihelion) $\beta_p = \frac{V_p}{c} = \beta\sqrt{\frac{1+e}{1-e}} = \frac{r_p\omega_o(0)}{c} = \frac{\kappa_p}{\sqrt{2}}\sqrt{1+e}$ (kinetic projection at perihelion)
$\delta_p = \frac{\kappa_p^2}{2\beta_p^2} = \frac{1}{1+e}$ (closure factor at perihelion)
$Q_p = \sqrt{\kappa_p^2 + \beta_p^2}$ (relational shift at perihelion)

Aphelion Relations

$r_a = a(1+e) = \frac{R_s}{\kappa_a^2}$ (radius at aphelion)
$\beta_a = \sqrt{\beta_p^2 e_X^{-2}} = \beta\sqrt{e_X^{-1}}$ (kinetic projection at aphelion)
$\kappa_a = \sqrt{\beta^2 + \beta_a^2}$ (potential projection at aphelion)
$\delta_\mathrm{a} = \frac{1}{1-e} = \frac{\kappa_a^2}{2\beta_a^2}$ (closure factor at aphelion)
$Q_a = \sqrt{\kappa_a^2 + \beta_a^2}$ (relational shift at aphelion)

Phase Variables (Depend on phases o and O)

$o = \frac{O}{\Omega}$ (idealised orbital phase in radians, represent phase o progression without orbital precession)
$O = \Omega \cdot o$ (true orbital phase with precession)
$r = r_o(O) = a\frac{1-e^2}{1+e\cos(O)} = \frac{R_s}{\kappa_o^2}$ (radial distance at phase O)
$O_r = \frac{1-e^2}{1+e\cdot\cos(O)} = \frac{r_o(O)}{a} = \frac{R_s}{\kappa_o(O)^2 a}$ (normalised true radial distance at phase O) $\kappa_o = \sqrt{\beta_R(O)^2 + \beta_T(O)^2 + \beta^2} = \sqrt{\frac{R_s}{r_o}} = \sqrt{1-(1+z_{\kappa o}(O))^{-2}} = \kappa_p\sqrt{\frac{1+e\cos(O)}{1+e}} = \sqrt{\beta^2+\beta_o^2} = \sqrt{\kappa_{surface}^2 \cdot \sin(\theta_o)}$ (local potential at the true phase O)
$\kappa_{Xo} = \sqrt{1-\kappa_o^2} = (z_{\kappa o}(O)+1)^{-1}$ (gravitational phase factor at phase O)
$\beta_o(O) = \beta\cdot\frac{\sqrt{1+e^2+2\cdot e\cdot\cos(O)}}{\sqrt{1-e^2}} = \sqrt{\kappa_o^2 - \beta^2} = \frac{\kappa_o(O)}{\sqrt{2\delta_o(O)}} = \sqrt{\frac{\kappa_o(O)^2}{2}\frac{1+e^2+2\cdot e\cdot\cos(O)}{1+e\cdot\cos(O)}} = \sqrt{\frac{R_s}{r_o(O)} - \frac{R_s}{2a}} = \sqrt{1-(1+z_{\beta o}(O))^{-2}}$ (kinetic projection at the true phase O)

$\beta_R(O) = \beta\ \frac{e\sin(O)}{\sqrt{1-e^2}} = \sqrt{\beta_o(O)^2 - \beta_T(O)^2} = \sqrt{((1-(1+z_{\kappa o}(O))^{-2}) - \beta^2) - \frac{r_o(O)^2\omega_o(O)^2}{c^2}}$ (radial kinetic projection at phase o)
$\beta_T(O) = \frac{r_o(O)\omega_o(O)}{c} = \beta\frac{1+e\cos(O)}{\sqrt{1-e^2}} = \kappa_o^2\frac{\sqrt{1-e^2}}{2\beta}$ (transverse kinetic projection at phase o)
$\beta_{Yo} = \sqrt{1-\beta_o^2} = (z_{\beta o}(O)+1)^{-1}$ (relativistic phase factor at phase O)
$\delta_o = \frac{1+e\cos o}{1+e^2+2e\cos o} = \frac{\kappa_o^2}{2\beta_o^2}$ (local closure factor at phase O)
$Q_o = \sqrt{\kappa_o^2+\beta_o^2}$ (local relational shift at phase O)
$\omega_o = a\beta c\frac{e_Y}{r_o^2} = \frac{\beta c}{a}\frac{(1+e\cos o)^2}{(1-e^2)^{3/2}}$ (angular speed at phase O)
$\eta_o = \frac{r}{a} = 2 - \frac{2\beta_o^2}{\kappa_o^2}$ (phase scale amplitude at phase)
$\tau_o = \kappa_{Xo}(O)\cdot\beta_{Yo}(O) = Z_{sys}(O)^{-1}$ (phase spacetime factor at phase O)
$\tau_{Yo} = \sqrt{1-\tau_o^2}$ (amplitude spacetime factor at phase)
$t_o = \frac{r_o}{c}$ (light time scale at phase)

Orbital Phase Invariants

Holographic Decryption Invariant

$$Z_{raw}(-\omega_i)\tau(-\omega_i)(1-e\cos\omega_i) + Z_{raw}(\pi-\omega_i)\tau(\pi-\omega_i)(1+e\cos\omega_i) = 2 \tag{1}$$

$\beta^2 = (\kappa_o^2 - \beta_o^2)$
$h_W = r_o^2\cdot\omega_o$
$\frac{\beta_T(O)}{\kappa_o^2(O)} = \frac{\sqrt{1-e^2}}{2\beta}$
$\beta_T(O)^2\eta_o^2 = \beta^2(1-e^2)$

Relational Geometry (WILL)

$\theta_1 = \arccos(\beta)$ (distribution angle on S^1, non-physical)
$\theta_2 = \arcsin(\kappa)$ (distribution angle on S^2, non-physical)
$\Delta_Q = Q_o^2 - Q^2$ (phase relational shift amplitude at phase O)
$B_a = \arccos(1-\delta^{-1}) = \arccos(-e) = \arccos(1-\frac{2\beta_p^2}{\kappa_p^2}) = \arccos(\frac{2\beta_a^2}{\kappa_a^2}-1)$ (ascending orbital balance point where $a=r$ and $\kappa_o^2 = 2\beta_o^2$ are true)
$B_d = 2\pi - \arccos(-e)$ (descending orbital balance point where $a=r$ and $\kappa_o^2 = 2\beta_o^2$ are true)

Structural-Dynamical Equivalence

$\frac{r_a}{r_p} = \frac{\delta_a}{\delta_p} = \frac{1+e}{1-e} = \frac{\beta_p}{\beta_a} = \frac{\kappa_p^2}{\kappa_a^2} = \frac{\kappa_a^2\beta_p^2}{\kappa_p^2\beta_a^2}$ (remarkable equivalence of structural (κ on S^2 carrier) and dynamical (β on S^1 carrier) symmetry suggesting once again that $SPACETIME \equiv ENERGY$)

Observer dependant

$\theta_{sys}(r) = \arcsin(\frac{R_{sys}}{r})$ (angular radius of central body at the distance r)
$Z_{raw}(O) = (1+\beta_{int}(\cos(O+\omega_i)+e\cos(\omega_i))\sin(i))\ Z_{sys}(O)$ (raw light shift including the line of site Doppler at phase o)
$\beta_z(O) = \frac{\beta}{\sqrt{1-e^2}}(\cos(O+\omega_i)+e\cos(\omega_i))\sin(i)$ (line of site light shift)
$i =$ (orbital inclination in relation to line of site and orbital plane)
$\omega_i =$ (phase turn or argument of periapsis)
$K_i = \frac{\beta}{\sqrt{1-e^2}}\sin(i) = \beta_{int}\sin(i)$ (semi-amplitude invariant)
$Z_{rawmax} = Z_{sys}(-\omega_i)(1+K_i(1+e\cos\omega_i))$
$Z_{rawmin} = Z_{sys}(\pi-\omega_i)(1+K_i(-1+e\cos\omega_i))$
$O_{oi} = (B_a+\omega_i)$
$Z_{sys}(-\omega_i) = (1-\beta^2\frac{1+e^2+2e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}(1-2\beta^2\frac{1+e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}$
$Z_{sys}(\pi-\omega_i) = (1-\beta^2\frac{1+e^2-2e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}(1-2\beta^2\frac{1-e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}$

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$$Z_{raw}(-\omega_i)\tau(-\omega_i)(1-e\cos\omega_i) + Z_{raw}(\pi-\omega_i)\tau(\pi-\omega_i)(1+e\cos\omega_i) = 2 \text{ (Holographic Decryption Invariant)}$$

Background Free Parametric Equations for the Observed Orbit

$$\begin{bmatrix} x_{sky} \\ y_{sky} \\ z_{depth} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(i) & -\sin(i) \\ 0 & \sin(i) & \cos(i) \end{bmatrix} \begin{bmatrix} x_{orb} \\ y_{orb} \\ 0 \end{bmatrix}$$

$$x_{sky} = r(O)\cos(O+\omega_i)$$
$$y_{sky} = r(O)\sin(O+\omega_i)\cos(i)$$
$$z_{depth} = r(O)\sin(O+\omega_i)\sin(i)$$

Un-tilted 2D coordinates within the orbital plane

$$x_{orb} = r(O)\cos(O+\omega_i)$$
$$y_{orb} = r(O)\sin(O+\omega_i)$$

3 S2 Star (Sgr A*) Data Alignment Comparison: R.O.M. vs GR

Instead of starting from the long chain of derivations and build our way to empirical testing, we will revers this order for our readers and will show the results first and then the way how we get them. By doing so we expect immediate engagement from the reader instead of the long and slow build up. Let us know if this upside down order succeed or fail in this little social experiment within this paper.
We apply the Relational Orbital Mechanics (R.O.M.) framework to the 24-year astrometric and radial-velocity dataset of the S2 star orbiting the Galactic Centre (Sgr A*). The dataset contains 174 raw observables (sky positions and line-of-sight velocities) and provides a sensitive test of the geometric closure condition $\kappa^2 = 2\beta^2$ in a strong-field regime.

Methodology

In the WILL RG model, the Keplerian orbital elements are supplemented by two relational features:

- The secular pericentre advance per orbital cycle is not a free parameter but is **algebraically locked** to the velocity amplitude β via the closure invariant:

$$f_{\text{prec}} = \frac{3\beta^2 - 2\beta^4}{1-e^2}, \qquad \beta = \frac{K}{c\,\sin i}\sqrt{1-e^2}. \tag{2}$$

- The relativistic contribution to the radial velocity is described by the *relational redshift*

$$Z_{\text{sys}} = \frac{c}{2}\left(\kappa_o^2 + \beta_o^2\right), \tag{3}$$

  where $\kappa_o^2 = R_s/r$ is the potential projection at the instantaneous orbital radius and $\beta_o^2 = \kappa_o^2 - \beta^2$ follows from the energy invariant.

The Rømer (light-travel time) delay is included as a single-step correction. The whole model contains **nine free parameters**: the angular semi-major axis a_{as}, eccentricity e, inclination i, argument of pericentre ω, longitude of ascending node Ω, orbital period P, epoch of pericentre passage T_0, systemic velocity V_0, and the radial-velocity semi-amplitude K.

For a fair comparison, we construct a General Relativity 1PN model with the **same nine parameters**. The pericentre advance is fixed by the standard 1PN formula

$$\Delta\omega_{\text{1PN}} = \frac{6\pi GM}{c^2 a(1-e^2)}, \tag{4}$$

which, once expressed through the Keplerian observables K, P, e, i, becomes a deterministic function of the fitted parameters with no additional degrees of freedom. The relativistic redshift in the GR model consists of the gravitational and transverse-Doppler terms. The Rømer delay is implemented identically in both models. The input data, error model, and optimisation algorithm are exactly the same for both fits.

System scale without G

In the WILL framework the Schwarzschild radius R_s is not obtained from a separately measured mass and Newton's constant, but is derived directly from the orbital period and the relational phase β:

$$R_s = \frac{T\,c\,\beta^3}{\pi}, \qquad a = \frac{R_s}{2\beta^2}. \tag{5}$$

Thus the entire physical scale of the system follows from dimensionless chronometric and spectroscopic ratios; G and the central mass appear only as *a posteriori* conversion factors to legacy units.

Results

Table 3 summarises the best-fit parameters and the fit quality for both models. The WILL RG model achieves a χ^2 of 727.20, compared to 800.81 for the GR 1PN model with the same number of free parameters.

Metric / Parameter	**GR 1PN (9 par.)**	**WILL RG (9 par.)**
Fit Quality (χ^2)	800.81	727.20
Relational Beta (β)	—	0.00642627
Eccentricity (e)	0.887826	0.887171
Inclination i (rad)	2.3707	2.3771
Period P (days)	5862.60	5862.49
Derived R_s (km)	—	1.2828×10^7
Derived Mass ($10^6\,M_\odot$)	4.24	4.34
Derived Distance R_0 (pc)	8275	8364
Free precession parameters	0	0
Computation time (Google Colab)	∼7 min	∼3 sec

Table 3: Comparison of the GR 1PN and WILL RG orbital solutions for S2. Both models have nine free parameters; the pericentre advance is fully determined by the fitted Keplerian elements in each case.

The improvement in χ^2 ($\Delta\chi^2 = 73.6$) is achieved without any additional adjustable parameters. The derived distance and central mass are consistent with recent independent measurements by the GRAVITY and Keck collaborations.

Google Colab Notebook

ROM vs GR S2 study.ipynb

Discussion

The WILL RG model provides a statistically better description of the S2 data than the standard 1PN parametrisation with the same parameter count. A detailed diagnostic decomposition (Section 3.1) reveals that the entire improvement is due to the phase-coupled treatment of the pericentre advance ($\omega_{\text{shift}} \propto \nu$), which replaces the usual time-linear advance. This parametrisation emerges naturally from the relational closure condition and is not an *ad hoc* modification.

The autonomous derivation of the system scale R_s directly from the orbital period and the dimensionless β illustrates that, for a purely gravitational two-body problem, the separation into G and M is unnecessary; the phenomenology is fully encoded in the geometric length R_s. This does not invalidate the use of G and mass in other domains of physics, but it demonstrates that orbital dynamics can be formulated in a self-contained, operationally minimal way.

Viewed as a purely kinematic refinement, the WILL parametrisation can be adopted within any relativistic orbit-fitting code without altering the underlying predictions of General Relativity. The computational speed (roughly 140 times faster than numerical integration of geodesic equations) makes it particularly suitable for large surveys and real-time monitoring.

3.1 Diagnostic: Decomposition of the Relational Orbital Model

In order to identify the precise origin of the substantial χ^2 improvement of the WILL RG fit over the standard GR 1PN fit (727.2 versus 800.8 for the S2 star), we performed a modular decomposition of the orbital model. The goal was to isolate which aspect of the relational framework – the precession law, the redshift formula, or the geometric parametrisation – is responsible for the better agreement with observations.

Methodology

We constructed a series of controlled numerical experiments using exactly the same data set, error model, and optimisation procedure as in the main S2 analysis (Section 3).

1. **Modular precession and redshift**: A generic model was built that can independently select
   - the precession advance rule: GR 1PN, GR 2PN, or WILL relational ($\tau_Y^2/(1-e^2)$);
   - the relativistic redshift formula: GR 1PN (gravitational + transverse Doppler) or WILL relational ($\frac{c}{2}(\kappa_o^2 + \beta_o^2)$).

   The number of free parameters remained nine in every combination, exactly as in the original fits.

2. **Discovery of the parametrisation mismatch**: All combinations that employed a standard *time-linear* advance of the argument of periapsis ($\omega(t) = \omega_0 + \dot{\omega}\,(t - T_0)$) gave $\chi^2 \approx 801$, regardless of whether the precession magnitude or the redshift were taken from GR or from WILL. Only the original WILL model, which uses a *true-anomaly-dependent* precession phase, reached $\chi^2 = 727$.

3. **Hybrid model**: To test whether the improved fit stems from the WILL precession *magnitude* or from the WILL *method of phasing the precession*, we built a hybrid that retains the exact geometric machinery of the WILL orbital model – in particular the phase shift $\omega_{\rm shift} = f_{\rm prec}\,\nu$ applied inside the radial equation and the Rømer delay – but replaces the WILL closure value $f_{\rm prec} = \tau_Y^2/(1-e^2)$ with the GR 1PN equivalent $f_{\rm prec} = 3\,(K/c)^2/\sin^2 i$.

Key Result

The hybrid model achieved

$$\chi^2_{\rm hybrid} = 727.20,$$

identical to the full WILL RG fit and significantly better than the standard GR 1PN fit ($\chi^2 = 800.81$). The precession magnitudes are numerically almost indistinguishable for S2:

$$f_{\rm prec}^{\rm WILL} = 0.00058183, \qquad f_{\rm prec}^{\rm GR\,1PN} = 0.00058185.$$

Thus the entire χ^2 improvement is due to the parametrisation of the precession, not its magnitude.

Interpretation: Phase-coupled versus time-linear precession

Standard GR implementation. In conventional post-Newtonian orbit fitting, the secular precession is modelled as a constant angular velocity:

$$\omega(t) = \omega_0 + \dot{\omega}\,(t - T_0), \qquad \dot{\omega} = \frac{\Delta\omega_{\rm 1PN}}{P}.$$

The true anomaly ν is computed from a Kepler equation that assumes the periapsis is fixed at the reference epoch T_0. Over an eccentric orbit, the true anomaly advances non-uniformly with time, so a constant $\dot{\omega}$ introduces a small but systematic phase error between the instantaneous orbital position and the intended precession state.

WILL parametrisation. The WILL RG model ties the precession directly to the accumulated true anomaly:

$$\omega_{\rm shift} = f_{\rm prec}\,\nu.$$

The instantaneous radius and the Rømer delay are evaluated with the shifted argument $(\nu - \omega_{\rm shift})$, which rigidly rotates the whole ellipse by an angle that is synchronised with the orbital motion. No separate time-linear advance of ω is imposed; the rotation is a function of the orbital phase itself. This coupling ensures that the pericentre direction is always correctly positioned for the instantaneous true anomaly.

Why the improvement is global. Because the substitution $t \to \nu$ in the precession phase affects the entire timing model, the correction is not localised near pericentre. It produces a slight but coherent shift of every predicted observables (astrometric positions and radial velocities) towards the measured values, exactly the uniform residual improvement observed in the diagnostic plots. Over the 24-year data set containing 174 observables, this small systematic adjustment accumulates to the $\Delta\chi^2 \approx -74$ seen in the fit.

Full MCMC Analysis

https://willrg.com/MCMC S2.html

The MCMC transforms a single optimizer result into a statistical statement: Across the full 9-dimensional parameter space, the ν-coupled precession parametrization fits 174 S2 observables better by $\Delta\chi^2 \approx 74$ than the time-linear parametrization, with no additional free parameters. This analysis show that this is not a fluke of the starting point or optimization algorithm.

Implications for Relational Geometry and for Standard Gravity

- **WILL RG is empirically preferred.** The relational closure condition $\kappa^2 = 2\beta^2$ naturally leads to the ν-dependent phase shift. No additional postulates, no knowledge of mass or G, and no background spacetime metric are required. These results indicate that the relational projection-closure approach naturally leads to an empirically preferred orbital model without requiring additional degrees of freedom.
- **Improved modeling tool.** Even within the framework of General Relativity, replacing the constant $\dot{\omega}$ with a ν-proportional advance (while keeping the same 1PN secular magnitude) is a more faithful realisation of the relativistic orbit. The present data strongly favour this re-parametrisation. Future high-precision monitoring of S2 and similar stars should adopt this method to obtain unbiased estimates of the central mass and distance.
- **Ontological parsimony.** WILL RG derives the same empirical performance without invoking G, M, spacetime curvature, or tensor calculus. It shows that all relevant observables can be extracted from the pure geometric closure of dimensionless projections ratios. It illustrate that for orbital dynamics the separation into G and M is unnecessary; the phenomenology is entirely encoded in the critical length (R_s). This does not rule out the necessity of G when connecting gravity to other domains of physics yet, but it clarifies the operational minimalism that Relational Geometry affords.

Summary of the Diagnostic Investigation

- The χ^2 advantage of the WILL RG model is caused entirely by its *phase-coupled* treatment of precession ($\omega_{\text{shift}} \propto \nu$), not by the specific value of the precession magnitude nor by the relational redshift alone.
- The standard GR time-linear implementation introduces a subtle but statistically significant systematic error in the modelled orbit.
- The WILL Relational Geometry automatically yields the superior parametrisation because it derives all dynamics from internal relations, avoiding the artificial separation of space and time and energy that leads to a time-linear approximation.

Conclusion

The modular analysis isolates the source of the improved fit to the WILL RG framework's phase-coupled treatment of pericentre advance. When the same GR 1PN secular advance is applied with a true-anomaly-dependent phasing, the model achieves the same χ^2 as the full WILL fit, significantly better than the standard time-linear implementation.

This indicates that the WILL-derived parametrisation – which follows directly from the closure of dimensionless relational projections – offers a more accurate empirical description of the S2 system. The analysis further demonstrates that, within the orbital domain, the critical scale R_s and all geometrical observables can be obtained without recourse to G or a kilogram mass.

These findings invite further investigation into Relational Geometry and its applications to the empirical structure of strong-field systems.

4 Relational Origin of:

4.1 Spectroscopic Shifts

Theorem 4.1 (Spectroscopic Phase Shift). *The geometric projection κ^2 of a closed system is strictly determined by the operationally measurable gravitational redshift z_κ via the identity:*

$$\kappa^2 = 1 - \frac{1}{(1+z_\kappa)^2}. \tag{6}$$

Proof. By the Conservation of Relation Theorem the gravitational amplitude κ and internal phase κ_X satisfy:

$$\kappa^2 + \kappa_X^2 = 1 \quad \Longrightarrow \quad \kappa^2 = 1 - \kappa_X^2. \tag{7}$$

By Relational Reciprocity Theorem, an observer must self-center at the relational origin $(\beta, \kappa) = (0, 0)$. Consequently, the observer's local phase is maximal: $\kappa_{X(obs)} \equiv 1$.

The observable spectroscopic shift z_κ is the operational measure of the discrepancy between the observer's internal phase and the source's internal phase. The optical scaling ratio is defined as:

$$1 + z_\kappa = \frac{\kappa_{X(obs)}}{\kappa_{X(source)}}. \tag{8}$$

Substituting the observer's maximal phase ($\kappa_{X(obs)} = 1$), the exact phase of the source is isolated:

$$\kappa_{X(source)} = \frac{1}{1 + z_\kappa}. \tag{9}$$

Substituting this internal phase back into the initial closure invariant yields the exact algebraic identity:

$$\kappa^2 = 1 - (\frac{1}{1 + z_\kappa})^2. \tag{10}$$

□

Theorem 4.2 (Kinematic Phase Shift (Transverse Doppler))**.** *The kinematic geometric projection β^2 of a closed system is strictly determined by the operationally measurable transverse Doppler shift z_β via the symmetric identity:*

$$\beta^2 = 1 - \frac{1}{(1 + z_\beta)^2}. \tag{11}$$

Proof. By the S^1 closure invariant (Conservation Theorem), the kinematic amplitude β and internal kinematic phase β_Y satisfy:

$$\beta^2 + \beta_Y^2 = 1 \quad \Longrightarrow \quad \beta^2 = 1 - \beta_Y^2. \tag{12}$$

Following the Principle of Relational Origin, the self-centered observer dictates the rest state, where external kinematic projection is zero ($\beta_{obs} = 0$), yielding maximum internal kinematic phase: $\beta_{Y(obs)} \equiv 1$.

The transverse Doppler shift z_β (measured orthogonally to eliminate classical radial kinematics) is the pure operational measure of the discrepancy between the observer's internal kinematic phase and the source's phase. The optical scaling ratio is:

$$1 + z_\beta = \frac{\beta_{Y(obs)}}{\beta_{Y(source)}}. \tag{13}$$

Substituting $\beta_{Y(obs)} = 1$, we isolate the internal phase of the moving source:

$$\beta_{Y(source)} = \frac{1}{1 + z_\beta}. \tag{14}$$

Substituting this back into the S^1 closure invariant yields the exact algebraic identity:

$$\beta^2 = 1 - (\frac{1}{1 + z_\beta})^2. \tag{15}$$

□

Theorem 4.3 (Operational Measurability)**.** *The relational projections are encoded directly in the combined phase interactions of light (spectroscopy) and are operationally independent of G and M.*

Proof. **Step 1: The Raw Observable (τ).** Spectroscopy does not measure "gravitational potential" or "kinematic velocity" as separate isolates; it measures the total accumulated phase difference between the source and the observer. We define the **Relational Spacetime Factor** τ as the inverse of the systems measured redshift product:

$\boxed{\kappa^2 = 1 - \frac{1}{(1 + z_\kappa)^2}}$ (z_κ = gravitational redshift)

$\boxed{\beta^2 = 1 - \frac{1}{(1 + z_\beta)^2}}$ (z_β = transverse Doppler shift)

$$\tau \equiv \frac{1}{Z_{sys}} = 1/[(1 + z_\kappa)(1 + z_\beta)]. \tag{16}$$

In the WILL framework, this single observable represents the product of the internal phase projections of the carriers S^2 and S^1:

$$\tau = \underbrace{\kappa_X}_{\text{Gravitational Phase}} \cdot \underbrace{\beta_Y}_{\text{Kinematic Phase}} = \sqrt{1-\kappa^2}\sqrt{1-\beta^2}. \tag{17}$$

Step 2: Exact Relational Identity. We reject weak-field approximations. The exact relationship between the signal τ and the structural norm Q is given by the algebraic expansion of the phase product:

$$\tau^2 = (1-\kappa^2)(1-\beta^2) = 1 - (\kappa^2+\beta^2) + \kappa^2\beta^2. \tag{18}$$

Substituting $Q^2 = \kappa^2 + \beta^2$, we obtain the rigorous link between the optical observable and the geometric state:

$$\tau^2 = 1 - Q^2 + \kappa^2\beta^2. \tag{19}$$

This identity demonstrates that the optical signal contains the complete information about the system's structural state, measurable without prior knowledge of mass. □

4.2 Kepler's Third Law

The Keplerian proportionality $a \propto T^{2/3}$ is a direct algebraic consequence of the energetic equilibrium between the 1-DOF directional kinematic carrier (S^1) and the 2-DOF omnidirectional potential carrier (S^2). The derivation strictly utilizes dimensionless ratios and absolute geometric scale, eliminating dependence on classical mass and the empirical gravitational constant.

The kinematic amplitude β on S^1 defines (sec:SR interval) the ratio of directional motion to the universal limit c:

$$\beta = \frac{2\pi a}{cT} \tag{20}$$

The omnidirectional potential amplitude squared κ^2 on S^2 is defined (sec:GR interval) by the absolute spatial geometric scale invariant R_s and the radial distance a:

$$\kappa^2 = \frac{R_s}{a} \tag{21}$$

Energetic closure dictates the invariant geometric exchange rate between the carriers:

$$\kappa^2 = 2\beta^2 \tag{22}$$

Substituting the parameterizations into the closure condition yields:

$$\frac{R_s}{a} = 2(\frac{2\pi a}{cT})^2 = \frac{8\pi^2 a^2}{c^2T^2} \tag{23}$$

Rearranging to isolate the semi-major axis a gives the exact closed-form relation:

$$a^3 = \frac{R_s c^2}{8\pi^2}T^2 \tag{24}$$

$$a = (\frac{R_s c^2}{8\pi^2})^{1/3} T^{2/3} \implies a \propto T^{2/3} \tag{25}$$

4.3 Eccentricity

Eccentricity is a measure of the projectional deviation from the circular equilibrium state ($\delta = 1$).

Theorem 4.4 (Geometric Eccentricity)**.** *For a closed orbital system governed by the projection invariants of WILL Relational Geometry, the orbital eccentricity e is strictly determined by the closure factor at periapsis, δ_p:*

$$e = \frac{2\beta_p^2}{\kappa_p^2} - 1 = \frac{1}{\delta_p} - 1. \tag{26}$$

Proof. Instead of relying on classical force laws, we derive this relation directly from the conservation of the two fundamental projection invariants of the WILL framework:

1. **Kinetic Projection Invariant (Binding Energy):** $\beta^2 = \kappa_o^2(o) - \beta_o^2(o) = \text{const}$.
2. **Angular Projection Invariant:** $h = r_o(o)\beta_T(o)c = \text{const}$ ($\beta = \beta_T(o)$ at turning points).

Consider the two turning points of a closed orbit: periapsis (p) and apoapsis (a). By operational definition of the shape parameter e, the relation between radii is determined by the geometric range:

$$r_a = r_p(\frac{1+e}{1-e}). \tag{27}$$

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Step 1: Relational Mapping. Using the angular invariant h (implying $\beta \propto 1/r$) and the field definition $\kappa^2 \propto 1/r$, we express the apoapsis projections in terms of the periapsis values:

$$\beta_a^2 = [\beta_p(\frac{r_p}{r_a})]^2 = \beta_p^2(\frac{1-e}{1+e})^2, \tag{28}$$

$$\kappa_a^2 = \kappa_p^2(\frac{r_p}{r_a}) = \kappa_p^2(\frac{1-e}{1+e}). \tag{29}$$

Note: Kinematic projection scales quadratically with the radius ratio, while potential projection scales linearly.

Step 2: Energy Balance. Substituting these into the kinetic global invariant β^2:

$$(\kappa_p^2 - \beta_p^2) = (\kappa_a^2 - \beta_a^2).$$

Substituting the mappings from Step 1:

$$\kappa_p^2 - \beta_p^2 = \kappa_p^2(\frac{1-e}{1+e}) - \beta_p^2(\frac{1-e}{1+e})^2.$$

Rearranging to group potential terms (κ) on the left and kinematic terms (β) on the right:

$$\kappa_p^2[1 - \frac{1-e}{1+e}] = \beta_p^2[1 - (\frac{1-e}{1+e})^2].$$

Step 3: Algebraic Reduction. Expanding the terms in brackets:

$$\text{LHS bracket } (\kappa \text{ term}): \quad 1 - \frac{1-e}{1+e} = \frac{(1+e)-(1-e)}{1+e} = \frac{2e}{1+e}.$$

$$\text{RHS bracket } (\beta \text{ term}): \quad 1 - \frac{(1-e)^2}{(1+e)^2} = \frac{(1+e)^2-(1-e)^2}{(1+e)^2} = \frac{4e}{(1+e)^2}.$$

Substituting back into the balance equation:

$$\kappa_p^2(\frac{2e}{1+e}) = \beta_p^2(\frac{4e}{(1+e)^2}).$$

Dividing both sides by $2e$ and multiplying by $(1+e)^2$:

$$\kappa_p^2(1+e) = 2\beta_p^2.$$

This yields geometric identity for bound orbits:

$$2\beta_p^2 = \kappa_p^2(1+e). \tag{30}$$

Step 4: Connection to Closure. Recall the definition of the closure factor at periapsis:

$$\delta_p = \frac{\kappa_p^2}{2\beta_p^2}.$$

Substituting Eq.(4.3) into this definition:

$$\delta_p = \frac{\kappa_p^2}{\kappa_p^2(1+e)} = \frac{1}{1+e}.$$

Solving for e, we obtain the stated result:

$$\boxed{e = \frac{1}{\delta_p} - 1 = \frac{2\beta_p^2}{\kappa_p^2} - 1 = 1 - \frac{2\beta_a^2}{\kappa_a^2}}$$

□

Remark 4.5. *This result confirms that eccentricity is strictly a measure of the energetic deviation from the circular equilibrium state ($\delta = 1$), derived entirely from the conservation of relational projections without invoking mass or Newtonian forces.*

SUMMARY

$$\frac{2\beta_p^2}{\kappa_p^2} - 1 \equiv e \equiv 1 - \frac{2\beta_a^2}{\kappa_a^2}$$

ECCENTRICITY ≡ CLOSURE DEFECT

SPACETIME ≡ ENERGY

4.4 Vis-Viva

In elliptical systems, the global closure theorem $\kappa^2 = 2\beta^2$ is conserved across the entire orbital cycle, while local phase variations manifest as an internal "breathing" of relational projections. We now derive the exact conservation law governing this local dynamic distribution.

Proposition 4.6 (Global Kinetic Amplitude)**.** *The difference between the square of the local potential projection $\kappa_o^2(o)$ and the square of the local kinetic projection $\beta_o^2(o)$ is a global binding invariant kinetic amplitude equal to β^2.*

Proof. The binding parameter β^2 defines the global relational depth of the system:

$$\kappa^2 = 2\beta^2$$

From the definitions of the S^2 and S^1 projections at any arbitrary phase O:

$$\kappa_o^2(o) = \frac{2\beta^2(1 + e\cos o)}{1 - e^2}$$

$$\beta_o^2(o) = \frac{\beta^2(1 + e^2 + 2e\cos o)}{1 - e^2}$$

Subtracting the local kinetic projection from the local potential projection:

$$\kappa_o^2(o) - \beta_o^2(o) = \frac{2\beta^2 + 2\beta^2 e\cos o - (\beta^2 + \beta^2 e^2 + 2\beta^2 e\cos o)}{1 - e^2}$$

$$\kappa_o^2(o) - \beta_o^2(o) = \frac{\beta^2(1 - e^2)}{1 - e^2} = \beta^2$$

Thus, the algebraic distance between the omnidirectional carrier S^2 and directional carrier S^1 relations remains globally constant irrespective of orbital phase and shape. □

Theorem 4.7 (The Orthogonal Signature of the Orbit)**.** *The square of the local potential projection $\kappa_o^2(o)$ on the S^2 carrier is the strict Pythagorean square sum of the local radial $\beta_R^2(o)$, transverse $\beta_T^2(o)$ and the global β^2 kinetic projections.*

Proof. The total local kinetic projection on the S^1 carrier splits orthogonally into radial and transverse components within the orbital plane:

$$\beta_o^2(o) = \beta_R^2(o) + \beta_T^2(o)$$

Substituting this orthogonal decomposition into the phase-invariant relation $\kappa_o^2(o) - \beta_o^2(o) = \beta^2$:

$$\kappa_o^2(o) - (\beta_R^2(o) + \beta_T^2(o)) = \beta^2$$

Rearranging yields the three-dimensional algebraic relational closure:

$$\boxed{\kappa_o^2(o) = \beta_R^2(o) + \beta_T^2(o) + \beta^2}$$

□

Remark 4.8 (Ontological Replacement of Vis-Viva)**.** *This geometric invariant replaces the descriptive Newtonian Vis-Viva equation ($v^2/2 - GM/r = -GM/2a$). Standard mechanics interprets this balance as a scalar subtraction of an abstract potential energy from kinetic energy. In WILL, it is revealed as a strict Pythagorean theorem of relational geometry: the local potential projection (κ_o^2) is generated by the orthogonal summation of the kinetic "breathing" ($\beta_R^2 + \beta_T^2$) and the global S^1 projection (β^2). SPACETIME ≡ ENERGY.*

4.5 Classical Acceleration

By the conservation of the global kinetic amplitude (Vis-Viva), the spatial gradients of the potential and kinetic projections balance to maintain the relational closure of the system. Taking the spatial derivative of the orbital invariant $\kappa_o^2 - \beta_o^2 = \beta^2$ with respect to the radial distance r yields:

$$\frac{d}{dr}(\kappa_o^2) = \frac{d}{dr}(\beta_o^2) \tag{31}$$

Substituting the definitions $\kappa_o^2 = \frac{R_s}{r}$ and $\beta_o^2 = \frac{v^2}{c^2}$ into the gradient balance:

$$\frac{d}{dr}\left(\frac{R_s}{r}\right) = \frac{1}{c^2}\frac{d(v^2)}{dr} \tag{32}$$

Evaluating the derivatives provides the geometric relationship between the potential scale and the kinematic spatial variation:

$$-\frac{R_s}{r^2} = \frac{2v}{c^2}\frac{dv}{dr} \tag{33}$$

Classical mechanics introduces the external time differential dt to evaluate this spatial variation. Applying the chain rule $a_{acc} = \frac{dv}{dt} = v\frac{dv}{dr}$ maps the kinematic spatial gradient to the classical acceleration (a_{acc}):

$$-\frac{R_s}{r^2} = \frac{2}{c^2}a_{acc} \tag{34}$$

Isolating a_{acc} yields the classical Newtonian formula:

$$a_{acc} = -\frac{R_s c^2}{2r^2} = -\frac{GM}{r^2} \tag{35}$$

Within the system, energy conservation is required by the closure theorem (sec:closure) as an invariant resource of state change projected on kinetic S^1 and potential S^2 carriers. Thus, there is no need to introduce any new primitives such as force or spacetime curvature. Geometric conservation of relational energy states provides a complete operational and ontological explanation of observed phenomena.

4.6 Angular Momentum Conservation

The conservation of specific angular momentum h is derived as a phase-independent structural invariant of the relational balance.

Proposition 4.9 (Invariant Ratio of Projections). *The transverse kinetic projection $\beta_T(o)$ is strictly proportional to the local potential projection $\kappa_o^2(o)$ scaled by the global system constants.*

Proof. From the definition of the local potential projection (based on closure theorem $\kappa^2 = 2\beta^2$ and geometry of ellipse):

$$\kappa_o^2(o) = 2\beta^2\frac{(1+e\cos o)}{1-e^2} \implies (1+e\cos o) = \kappa_o^2(o)\frac{1-e^2}{2\beta^2}$$

The transverse kinetic projection $\beta_T(o)$ is defined on the S^1 carrier as:

$$\beta_T(o) = \beta\frac{1+e\cos o}{\sqrt{1-e^2}}$$

Substituting the expression for $(1+e\cos o)$ into the definition of $\beta_T(o)$:

$$\beta_T(o) = \beta(\kappa_o^2(o)\frac{1-e^2}{2\beta^2})\frac{1}{\sqrt{1-e^2}} = \kappa_o^2(o)\frac{\sqrt{1-e^2}}{2\beta}$$

Rearranging to isolate the invariant ratio of the phase-dependent projections:

$$\frac{\beta_T(o)}{\kappa_o^2(o)} = \frac{\sqrt{1-e^2}}{2\beta}$$

□

Theorem 4.10 (Conservation of Angular Momentum). *The specific angular momentum h, defined as the product of the local scale $r_o(o)$ and the transverse kinetic projection, is a global phase invariant.*

Proof. Using the local scale $r_o(o) = R_s/\kappa_o^2(o)$, we define h:

$$h = r_o(o)\beta_T(o)c = \frac{R_s}{\kappa_o^2(o)}\beta_T(o)c$$

Substituting the invariant ratio $\frac{\beta_T(o)}{\kappa_o^2(o)} = \frac{\sqrt{1-e^2}}{2\beta}$:

$$\boxed{h = R_s c\frac{\sqrt{1-e^2}}{2\beta}}$$

Since R_s, e, and β are global invariants, h does not depend on phase o. Angular momentum is thus the manifestation of the fixed structural exchange rate between the S^2 and the S^1 carriers and there projections. □

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4.7 Orbital Precession as Phase Accumulation

In standard General Relativity, orbital precession is derived via perturbative methods and Taylor expansions of the Schwarzschild metric. In WILL Relational Geometry, precession is derived strictly from the exact algebraic accumulation of the internal phase difference over a closed geometric cycle, without differential approximations.

Theorem 4.11 (Precession Law)**.** *The secular angular shift (precession) $\Delta\varphi$ per closed orbit is the strict geometric projection of the system's relational phase difference τ_Y normalized by the orbital shape factor $(1 - e^2)$.*

Proof. **Step 1: State Difference.** A system completely at rest with the observer possesses maximal internal phase $\tau = 1$. The true relational phase of an orbiting system is the product of the kinetic S^1 and potential S^2 projections carriers:

$$\tau^2 = \beta_Y^2 \kappa_X^2 = (1 - \beta^2)(1 - \kappa^2). \tag{36}$$

Expanding this product algebraically yields:

$$\tau^2 = 1 - (\beta^2 + \kappa^2) + \kappa^2\beta^2. \tag{37}$$

Recognizing the frame invariant scale of the relational shift $Q^2 = \beta^2 + \kappa^2$; The total phase divergence from the rest origin ($\tau = 1$) is defined as:

$$\tau_Y^2 \equiv 1 - \tau^2 = Q^2 - \kappa^2\beta^2. \tag{38}$$

The cross-coupling term $\kappa^2\beta^2$ account for non-linear geometric interaction between the directional S^1 and omnidirectional S^2 relational carriers.

Step 2: Geometric Accumulation of Phase Divergence. The system accumulates this state mismatch over the full cycle (2π). The total angular shift is the base phase scaled by the state divergence, normalized by the orbital shape factor $(1 - e^2)$.

Restricted by the Mathematical Transparency principle (pr:mathematical), we do not truncate or simplify this connection. The orbital precession is locked in its exact, fully non-linear algebraic form:

$$\boxed{\Delta\varphi = \frac{2\pi \cdot \tau_Y^2}{1 - e^2} = \frac{2\pi(1 - \tau^2)}{1 - e^2}} \tag{39}$$

□

Remark 4.12 (Epistemic Purity)**.** *This derivation contrast with the complex integration of perturbed differential equations reviling the exact algebraic evaluation of the relational phase. The cross-coupling term $\kappa^2\beta^2$ distinct RG from just approximation and structurally complete the exact invariant topology (full mathematical prove 5).*

4.8 Dynamic Event Horizon

In classical literature, the event horizon is often heuristically treated as an escape-velocity boundary where spacetime curvature becomes infinitely steep. In RG, the horizon emerges strictly as the limit of geometric precessional invariant, representing the ultimate topological closure of a trajectory.

Using the exact relational phase accumulation law (derived in 4.7), the baseline precession per orbit is given by:

$$\Delta\varphi = \frac{2\pi \cdot \tau_Y^2}{1 - e^2} = \frac{2\pi(1 - \tau^2)}{1 - e^2} \tag{40}$$

For a massive system under total relational closure ($\kappa^2 = 2\beta^2$), consider the **Dynamic Horizon** state where the potential projection reaches absolute saturation ($\kappa^2 = 1$). By the closure rule, the kinematic projection is $\beta^2 = \frac{1}{2}$.

At this limit, the gravitational phase component collapses ($\kappa_X = \sqrt{1 - 1} = 0$), causing the entire internal causal structure of the system to vanish:

$$\tau^2 = (1 - \beta^2)(1 - \kappa^2) = 0 \implies \tau_Y = 1 \tag{41}$$

For a perfectly circular orbit ($e = 0$) at this causal limit, the precession evaluates to:

$$\Delta\varphi = \frac{2\pi \cdot (1)}{1 - 0} = 2\pi \tag{42}$$

Event Horizon

An orbital precession of $\Delta\varphi = 2\pi$ radians per revolution implies that the periapsis shifts by a full circle during a single orbit. Consequently, the orbital progression continuously folds onto itself perfectly, creating a closed, self-intersecting locus where forward orbital motion mathematically nullifies itself. This provides a novel interpretation of the event horizon phenomena not as a "point of infinite density" in an external container-like spacetime, but the unitary topological closure of the relational trajectory itself.

4.9 Gravitational Deflection and Lensing

In General Relativity, the deflection of light is obtained by integrating the null geodesic equations over a curved spacetime manifold, often relying on weak-field approximations and Taylor expansions. Within WILL Relational Geometry (RG), we reject both the background manifold and the use of mathematical approximations as non-operational ontological artifacts.

The system consists exclusively of its participants: the Source, the Lens (at periapsis p), and the Receiver. The total deflection angle must be derived as a strict, exact algebraic difference between their measurable relational phase states, without resorting to series expansions.

Gravitational Deflection and Interaction Gradient

Energy projections distribution among axis of relational carriers has to obey the conservation law (distribution between axis does not create nor destroy energy). There must exist a strict, algebraically closed gradient connecting all states, governed entirely by the kinematic projection β.

Theorem 4.13 (Unified Interaction Gradient)**.** *The gravitational interaction capacity of any entity is determined by its available phase buffer β_Y. The geometric scaling factor Γ that dictates the distribution of the potential projection κ^2 onto the spatial trajectory is strictly defined by the arithmetic mean of the saturated carrier S^1:*

$$\Gamma(\beta) = \frac{1+\beta^2}{2} = 1 - \frac{\beta_Y^2}{2} \tag{43}$$

Proof. By the S^1 closure invariant ($\beta^2 + \beta_Y^2 = 1$), an object at rest ($\beta = 0$) possesses maximum internal phase ($\beta_Y = 1$). This phase acts as a geometric buffer, absorbing half of the relational gradient, yielding the classical partitioning $\Gamma = \frac{1}{2}$. As the spatial projection β increases, the internal clock slows, and the phase buffer β_Y depletes. At the topological limit $\beta \to 1$ (light), the internal phase collapses ($\beta_Y \to 0$). The buffer is exhausted, forcing the entity to absorb the full, unpartitioned gravitational gradient, yielding $\Gamma = 1$. □

Using this gradient, we define the **Unified Closure Defect** for any trajectory—from a slow asteroid to a photon—as:

$$\delta_\varphi = \frac{\kappa_p^2}{\beta_p^2}\Gamma(\beta_p) = \frac{\kappa_p^2(1+\beta_p^2)}{2\beta_p^2} \tag{44}$$

The geometric shape parameter (eccentricity) of the trajectory is derived directly from this defect:

$$e_\varphi = \frac{1}{\delta_\varphi} - 1 = \frac{2\beta_p^2}{\kappa_p^2(1+\beta_p^2)} - 1 \tag{45}$$

Applying the exact algebraic transit equation for a distant observer ($\kappa_o \to 0$), we have $\cos o_\infty = -1/e_\varphi$. Using the same trigonometric extraction as established for light ($\sin(\frac{\Delta\varphi}{2}) = \frac{1}{e_\varphi}$), we arrive at the absolute, unified equation for gravitational deflection:

$$\boxed{\Delta\varphi = 2\arcsin(\frac{\kappa_p^2(1+\beta_p^2)}{2\beta_p^2 - \kappa_p^2(1+\beta_p^2)})} \tag{46}$$

Solving for $\beta_p = 1$ the total deflection angle $\Delta\varphi$ gives non-linear equation for light deflection in RG:

$$\boxed{\Delta\gamma = 2\arcsin(\frac{\kappa_p^2}{\kappa_{Xp}^2})} \tag{47}$$

Desmos and Colab

Desmos: Algebraic Light Deflection (one input derivation) Colab: Gravitational Deflection.ipynb

Verification of Topological Limits:

- **Newtonian Limit ($\beta_p \ll 1$):** The phase buffer is full ($\Gamma \to 0.5$). The eccentricity is dominated by $2\beta_p^2/\kappa_p^2$. The deflection reduces to the classical Rutherford/Newton scattering: $\Delta\varphi \approx \frac{\kappa_p^2}{\beta_p^2}$.
- **Relativistic Limit ($\beta_p \to 0.99$):** The phase buffer is nearly depleted ($\Gamma \to 0.99$). The trajectory stiffens, and the deflection angle approaches the photonic maximum, smoothly capturing the post-Newtonian factor without Taylor expansions.

- **Photonic Limit ($\beta_p = 1$):** The phase buffer is completely exhausted ($\Gamma = 1$). Substituting $\beta_p = 1$ yields $e = \frac{2}{\kappa_p^2(2)} - 1 = \frac{1}{\kappa_p^2} - 1 = \frac{\kappa_{Xp}^2}{\kappa_p^2}$. The equation resolves perfectly into the exact light deflection identity: $\Delta\varphi = 2\arcsin(\frac{\kappa_p^2}{\kappa_{Xp}^2})$.

Summary

This formulation suggests that light and massive bodies do not require separate ontological categories or distinct dynamical laws. In RG, the photonic state the topological limit ($\beta \to 1$, $\beta_Y \to 0$) of the phase buffer. The historical factor of 2 in gravitational lensing might not be an anomaly of null geodesics in curved background spacetime, but the inevitable geometric consequence of this exhausted phase capacity."

Gravitational Lensing and the Einstein Ring

In Relational Geometry, the description of gravitational lensing (deflection, convergence, shear, and magnification) does not require a background metric or the postulation of a curved manifold. It arises entirely as the geometric projection of the deflection angle onto the observer's plane.

By employing the standard geometric baseline equation for the observer-lens-source system, the true angular position θ_S of a source is related to its observed image position θ_I via:

$$\theta_S = \theta_I - \frac{D_{LS}}{D_S}\Delta\varphi_{\text{unified}}(\beta_p, \kappa_p(\theta_I)) \tag{48}$$

where D_{LS} and D_S are the relational baseline distances.

The potential projection at periapsis κ_p^2 is strictly a function of the observed angle θ_I, defined by the physical impact parameter $r_p = D_L\theta_I$:

$$\kappa_p^2(\theta_I) = \frac{R_s}{D_L\theta_I} \tag{49}$$

Theorem 4.14 (Exact Algebraic Einstein Ring)**.** *For a perfectly aligned system ($\theta_S = 0$), the image forms a symmetric ring. The exact angular radius of this Einstein ring, θ_E, valid for all velocities and field strengths, is given by the implicit algebraic equation:*

$$\theta_E = 2\frac{D_{LS}}{D_S}\arcsin(\frac{\kappa_p^2(\theta_E)(1+\beta_p^2)}{2\beta_p^2 - \kappa_p^2(\theta_E)(1+\beta_p^2)}) \tag{50}$$

Recovery of Topological Limits: To reveal the underlying gradient without altering the exact equation, we examine the weak-field astrophysical limit ($\kappa_p^2 \ll \beta_p^2$), where $\arcsin(x) \approx x$. The unified deflection simplifies algebraically to:

$$\theta_E \approx \frac{D_{LS}}{D_S}\frac{R_s}{D_L\theta_E}(\frac{1+\beta_p^2}{\beta_p^2}) \quad \Longrightarrow \quad \theta_E^2 \approx \frac{R_sD_{LS}}{D_LD_S}(\frac{1+\beta_p^2}{\beta_p^2}) \tag{51}$$

Colab Notebook

Gravitational Lensing.ipynb

- **The Photonic Limit ($\beta_p \to 1$):** The kinematic factor evaluates exactly to $\frac{1+1}{1} = 2$. The equation resolves to $\theta_{E,\gamma} = \sqrt{\frac{2R_sD_{LS}}{D_LD_S}}$, perfectly matching the full General Relativistic prediction for light (equivalent to $4GM/c^2$).
- **The Relativistic Massive Limit ($0 < \beta_p < 1$):** The continuous phase-buffer depletion natively supplies the correct post-Newtonian scaling without external ad-hoc gradients.

Epistemological Consequence

The properties of the lens - convergence (κ_{lens}), shear (γ), and magnification (μ) - are the components of the 2D Jacobian matrix of the deflection field $\vec{\alpha}(\vec{\theta})$. Because the physics of the phase-buffer β_Y is fundamentally woven into the exact definition of $\Delta\varphi_{\text{unified}}$, no manual "weighting" of shear or convergence is required. The optical mapping emerges naturally from the pure algebraic closure of the S^1 and S^2 relational carriers.

5 Mercury's Precession: GR 1PN as First-Order Approximation of RG

5.1 Statement of the Verification Problem

The standard first-order post-Newtonian expression for the perihelion advance of a bound two-body orbit is

$$\Delta\varphi_{\rm GR} \;=\; \frac{3\pi\, R_s}{a\,(1-e^2)}, \tag{52}$$

with a the semi-major axis, e the eccentricity, and $R_s = 2GM/c^2$ the Schwarzschild scale of the central body.

WILL Relational Geometry (WILL RG) expresses the secular angular shift per closed orbit as the strict geometric accumulation of the system's phase divergence from the rest origin:

$$\Delta\varphi_{\rm RG} \;=\; \frac{2\pi\,\tau_Y^2}{1-e^2}, \qquad \tau_Y^2 \;\equiv\; 1-\tau^2 \;=\; 1-(1-\beta^2)(1-\kappa^2) \;=\; \beta^2+\kappa^2-\beta^2\kappa^2, \tag{53}$$

evaluated at the orbital reference scale (semi-major axis a), where the closure condition $\beta^2(a) = \kappa^2(a)/2$ and the field relation $\kappa^2(a) = R_s/a$ hold.

The question addressed in this section is whether (52) is the leading-order content of (53) in the small parameter R_s/a. The structure of (53) suggests an even stronger statement than in the time-dilation case (sec:earth-gps): the GR formula should be recoverable by removing a single explicit term — the cross-coupling $\beta^2\kappa^2$ — from τ_Y^2, with the residual being exactly $-2\pi\,\beta^2\kappa^2/(1-e^2)$.

5.2 Methodology

The verification is split into two independent parts, both executed in a single public notebook (Section 5.5).

Part I — symbolic identity. The exact WILL RG precession is expressed in closed form by substituting the closure condition $\beta^2 = \kappa^2/2$ and $\kappa^2 = R_s/a$ into (53). The result is simplified algebraically and subtracted from (52). The algebraic identity between the residual and the analytically predicted second-order term $-2\pi\,\beta^2\kappa^2/(1-e^2)\big|_a$ is verified using `sympy.simplify`. As an independent cross-check, (53) is Taylor-expanded in a bookkeeping parameter ε that scales R_s, and the first-order coefficient is compared with (52).

Part II — numerical comparison. Both (52) and (53) are evaluated with 40-digit precision arithmetic (`mpmath`, `mp.dps = 40`) using Mercury's J2000 orbital elements: $a = 5.7909050 \times 10^{10}\,$m, $e = 0.20563593$, orbital period $T = 87.9691\,$d, $GM_\odot = 1.32712440018 \times 10^{20}\,\mathrm{m^3\,s^{-2}}$, $c = 299{,}792{,}458\,\mathrm{m\,s^{-1}}$, giving $R_s^\odot = 2953.25\,$m. Per-orbit values are converted to arcseconds per Julian century using $36{,}525\,\mathrm{d}/T$ orbits per century. The numerical discrepancy $\Delta\varphi_{\rm RG} - \Delta\varphi_{\rm GR}$ is compared with the analytically predicted second-order correction obtained in Part I.

5.3 Results

Symbolic. Substituting the closure and field relations into (53) yields the closed-form WILL RG precession

$$\Delta\varphi_{\rm RG} \;=\; \frac{3\pi\, R_s}{a\,(1-e^2)} \;-\; \frac{\pi\, R_s^2}{a^2\,(1-e^2)}. \tag{54}$$

The first term is identically (52). The algebraic difference is therefore

$$\Delta\varphi_{\rm RG} - \Delta\varphi_{\rm GR} \;=\; -\,\frac{\pi\, R_s^2}{a^2\,(1-e^2)} \;\equiv\; -\,\frac{2\pi\,\beta^2\kappa^2}{1-e^2}\bigg|_a. \tag{55}$$

The identity $\big[\Delta\varphi_{\rm RG} - \Delta\varphi_{\rm GR}\big] - \big[-2\pi\,\beta^2\kappa^2/(1-e^2)\big]\big|_a = 0$ vanishes identically in (R_s, a, e). The Taylor expansion of (53) in R_s/a reproduces (52) at first order with zero residual.

Structural observation. In the time-dilation case the GR additive formula was the linear coefficient of a Taylor expansion of a multiplicative τ-ratio. In the present case the correspondence is more direct: the GR first-order PN precession formula coincides exactly with the WILL RG expression *after a single explicit cancellation*, namely $\beta^2\kappa^2 \to 0$ in $\tau_Y^2 = 1-(1-\beta^2)(1-\kappa^2)$. The omitted term encodes the second-order coupling between the kinematic S^1 and gravitational S^2 projection carriers of the WILL framework.

Numerical. With Mercury's parameters listed above:

quantity	value
$\Delta\varphi_{\text{RG}}$ (exact, per orbit)	$5.01867565337 \times 10^{-7}$ rad
$\Delta\varphi_{\text{GR}}$ (first-order PN, per orbit)	$5.01867573869 \times 10^{-7}$ rad
$\Delta\varphi_{\text{RG}} - \Delta\varphi_{\text{GR}}$ (per orbit)	-8.531×10^{-15} rad
$-\pi R_s^2/[a^2(1-e^2)]$ (predicted, per orbit)	-8.531×10^{-15} rad
residual	$\sim 10^{-41}$ rad
$\Delta\varphi_{\text{RG}}$ (per century)	$42.9807844914''$
$\Delta\varphi_{\text{GR}}$ (per century)	$42.9807852220''$
$\Delta\varphi_{\text{RG}} - \Delta\varphi_{\text{GR}}$ (per century)	$-7.306 \times 10^{-7}\,''$

The numerical discrepancy $\Delta\varphi_{\text{RG}} - \Delta\varphi_{\text{GR}}$ matches the analytically derived second-order term (55) to within the limit set by the 40-digit floating-point arithmetic. The relative magnitude of the term omitted by the first-order GR formula is $\sim 1.7 \times 10^{-8}$ of the total precession.

5.4 Interpretation

Equation (54) establishes that the standard first-order post-Newtonian formula (52) is the leading-order content of the WILL RG exact expression (53) in R_s/a, and that the omitted contribution is exactly the cross-coupling term $-2\pi\,\beta^2\kappa^2/(1-e^2)$ at the semi-major axis. This is an algebraic identity in (R_s, a, e), not a numerical near-match.

For Mercury, the second-order term in (54) contributes at the level of $-7.3\times10^{-7}\,''\,\text{century}^{-1}$, i.e. $\sim 1.7\times10^{-8}$ of the total precession of $\approx 42.98''\,\text{century}^{-1}$. This deviation is well below the current precision of perihelion measurements and does not, by itself, constitute a distinguishing observation between the two formulations in the Solar System regime. The sign of the correction is fixed: WILL RG predicts a slightly smaller precession than the first-order post-Newtonian formula.

The scope of the present statement is restricted to the relation between the WILL RG closed-form expression (53) and the *first-order post-Newtonian* GR precession formula (52). A comparison against fully non-linear geodesic integration in the exact Schwarzschild geometry — which itself receives positive-definite corrections beyond the leading $3\pi R_s/[a(1-e^2)]$ term — is a separate calculation and is not addressed in this section.

5.5 Reproducibility

The complete symbolic and numerical verification reported in this section is contained in a single public Google Colab notebook, comprising the two scripts used to produce (54), (55), and the numerical table above:

Colab Notebook

Mercury precession RG 1st order=GR.ipynb

The notebook has no external dependencies beyond `sympy` and `mpmath` and is self-contained. All input parameters are listed explicitly in the numerical script; any change in a, e, or $GM_\odot$ rescales both $\Delta\varphi_{\text{RG}}$ and $\Delta\varphi_{\text{GR}}$ consistently and does not affect the symbolic identity (55).

6 Relational Parameterization and the Fundamental Primitives

Theorem 6.1 (Relational Parameterization)**.** *The complete structural and dynamical parameterization of any closed orbital system - specifically its geometric shape (e - eccentricity), secular phase shift ($\Delta\varphi$ - orbital precession), critical scale (R_s - Schwarzschild radius), and global scale (a - semi-major axis) - are strictly determined by the algebraic relations of its dimensionless spectroscopic and chronometric observables. The derivation requires no independent mass parameter (M), gravitational constant (G), or a priori spatial metric and deferential or tensor formalisms.*

Direct algebraic derivation via spectroscopic and chronometric ratios.

Proof. The proof is constructive and algebraically closed. It generalizes to any bound orbital system. We demonstrate this using the Mercury-Sun system as an operational test case and Jupiter-Sun as confirmation test. All governing equations are derived exclusively from the internal topology of the WILL relational carriers (S^1, S^2), strictly bypassing Newtonian mechanics, Special Relativity, or General Relativity. The derivation is executed using only four dimensionless, epistemologically direct observables $(e, \theta, T_{ratio}, z_{sun})$:

INPUTS (Cross-Cultural Invariants)

The following inputs are strict dimensionless ratios. They represent direct physical observables that remain identical for any observer in the Universe, regardless of local unit systems (meters/seconds) or theoretical paradigms.

Parameter	Value	Direct Observational Method
	Sun's physical observables	
$\theta_\odot$	0.0046524 rad	**Visual Size:** The angular radius of the central body in the sky. [?]
z_{sun}	2.1224×10^{-6}	**Spectroscopic Tension:** The raw fractional frequency shift of light $(1-\nu_{obs}/\nu_{emit})$. Defined strictly as an optical geometric observable, independent of mass or Newtonian constants [?].
	Mercurye's physical observables	
$T_{\text{ratioM}} = T_M/T_\oplus$	0.2408	**Cycle Ratio:** The observed system's full orbital period divided by the observer's local orbital period. [?]
e_M	0.2056	**Visual Kinematics:** Derived directly from the ratio of the planet's fastest (ω_{max}) and slowest (ω_{min}) apparent angular speeds: $e = \frac{\sqrt{\omega_{max}/\omega_{min}}-1}{\sqrt{\omega_{max}/\omega_{min}}+1}$ [?].
	Jupiter's physical observables	
$T_{\text{ratio}J} = T_J/T_\oplus$	11.862	**Cycle Ratio:** The full Jovian orbital period measured against the observer's local (Earth) orbital period — a purely temporal, clock-based observable requiring no unit conversion or gravitational constant [?].
e_J	0.04839266	**Visual Kinematics:** Derived directly from the ratio of Jupiter's fastest (ω_{max}) and slowest (ω_{min}) apparent angular speeds along its orbit: $e = \frac{\sqrt{\omega_{max}/\omega_{min}}-1}{\sqrt{\omega_{max}/\omega_{min}}+1}$ [?].

Algebraic Derivation

1. Relational Scale Factor

The ratio of the Sun's radius to the planet's perihelion distance is derived entirely from visual size and cycle ratios:

$$R_{ratio} = \frac{\theta_\odot}{(T_M/T_\oplus)^{2/3} \cdot (1-e_M)} \approx 0.0151290$$

2. Potential Projection (at perihelion)

Translating the spectral shift at the Sun's surface (z_{sun}) to the potential projection (4.1) at Mercury's perihelion (κ_p^2) using the derived scale factor:

$$\kappa_p^2 = (1 - \frac{1}{(1+z_{sun})^2}) \cdot R_{ratio} \approx 6.4219 \times 10^{-8}$$

3. Kinematic Projection (at perihelion)

Deriving orbital kinematics strictly from the potential and eccentricity (4.3), requiring no Doppler or radar data:

$$\beta_p^2 = \frac{\kappa_p^2}{2}(1+e_M) \approx 3.8712 \times 10^{-8}$$

4. Global Orbital Invariants

Extracting the system's global kinetic projection (β) via the invariant binding energy $\beta^2 = \kappa_p^2 - \beta_p^2$ (via Energy-Symmetry Law sec:energy-symmetry):

$$\beta^2 =\approx 2.5508 \times 10^{-8} \quad \text{and} \quad \kappa^2 = 2\beta^2 \approx 5.1016 \times 10^{-8}$$

(via Closure theorem thm:closure).

5. Exact Orbital Precession

The secular phase shift emerges (4.7) from the accumulation of the relational state divergence ($\tau_Y^2 = 1-(1-\kappa^2)(1-\beta^2)$) over a full cycle:

$$\Delta\varphi = \frac{2\pi \cdot \tau_Y^2}{1-e_M^2} \approx 5.0203 \times 10^{-7} \text{ rad/orbit}$$

Converting this pure geometric phase shift to standard observational units yields **43.015 arcsec/century**, perfectly matching the observed precession of Mercury.

6. Absolute System Scale (Metric Translation)

To translate the geometric scale into legacy metric units (meters), we map Mercury's orbital period ($T_M = 7,600,521.6$ s) and the speed of light (c) to the global kinematic projection (β):

$$R_{sRG} = T_M \cdot c \cdot \frac{\beta^3}{\pi} \approx 2954.8 \text{ m}$$

Standard GR value:

$$R_{sGR} = \frac{2GM_{sun}}{c^2} \approx 2953.33 \text{ m} \quad \text{where} \quad M_{sun} = 1.98847 \times 10^{30} \text{ kg}$$

Discrepancy:

$$\frac{R_{sGR} - R_{sRG}}{R_{sGR}} \times 100 = |0.0476|\%$$

This variance is strictly within the observational uncertainty limits of the input parameters (z_{sun} and $\theta_\odot$).

7. Semi-major Axis (Standard value $\approx 5.79 \times 10^{10}$ m):

$$a = \frac{R_{sRG}}{\kappa^2} \approx 5.792288 \times 10^{10} \text{ m}$$

8. Perihelion Radius (Standard value $\approx 4.60 \times 10^{10}$ m):

$$r_p = \frac{R_{sRG}}{\kappa_p^2} \approx 4.601394 \times 10^{10} \text{ m}$$

Universality of the Framework: The Jovian Extension

To rigorously demonstrate that this parameterization is not a localized artifact of the Mercury-Sun proximity, but a universal geometric property of bound systems, we apply the exact same algebraic chain to Jupiter. We replace the inputs with Jupiter to Earth orbital periods ratios (T_{ratioJ}) and Jupiter's eccentricity (e_J), keeping the central reference invariants ($\theta_\odot$, z_{sun}, $T_\oplus$) strictly identical.

Jovian Inputs:

- $T_J = 4332.589$ days [?]
- $e_J = 0.04839266$ [?]
- $T_{ratioJ} = T_J/T_\oplus \approx 11.862$ [?]

1. Global Potential Projection (κ_J)

The local potential projection at Jupiter's semi-major axis reduces to a remarkably elegant relation, scaling the solar surface tension strictly through the inverse temporal cycle ratio:

$$\kappa_J = \sqrt{(1 - \frac{1}{(1+z_{sun})^2}) \cdot \frac{\theta_\odot}{(T_{ratioJ})^{2/3}}} \approx 6.1623 \times 10^{-5}$$

2. Absolute System Scale Recovery (R_{sRGJ})

Expanding the algebraic chain utilizing Jupiter's specific kinematic projection (β_J) yields the invariant core scale:

$$R_{sRGJ} = \frac{T_J \cdot c}{\pi}[\frac{1}{1-e_J}(1 - \frac{1}{(1+z_{sun})^2})\frac{\theta_\odot}{T_{ratioJ}^{2/3}} - \frac{1}{2}\frac{1+e_J}{1-e_J}(1 - \frac{1}{(1+z_{sun})^2})\frac{\theta_\odot}{T_{ratioJ}^{2/3}}]^{3/2}$$

$$R_{sRGJ} \approx 2955.41 \text{ m}$$

Jovian Orbital Recovery

Translating the recovered systemic scale to the spatial scale of Jupiter's orbit:

$$a_J = \frac{R_{sRGJ}}{\kappa_J^2} \approx 7.7827 \times 10^{11} \text{ m}$$

Observed value $a_{JDATA} = 778.57 \times 10^9$ m.

$$\text{Discrepancy} = \frac{|a_{JDATA} - a_J|}{a_{JDATA}} \times 100 \approx 0.038\%$$

The reconstruction of the Schwarzschild radius (R_s) and the orbital scale (a) from disparate chronometric cycles (T, e) demonstrates that there's a constant direct invariant relation between:

- **potential** geometry (κ^2 - **"spacetime structure"**)
- and system's **kinetic** (β^2- **"energy dynamics"**) both seen as relational measure of state difference, rooting directly from foundational unification principle **SPACETIME ≡ ENERGY** (pr:unifying).

□

Conclusion

All major parameters of gravitational systems are recovered through algebraic closure without mass (M), gravitational constant (G), metric manifold, differential formalism, or external temporal flow.

Corollary 6.2 (Epistemic Mandate and Parameter Parsimony)**.** *If a system's generative chain is algebraically complete without a specific parameter, that parameter is an emergent scaling constant rather than a fundamental primitive. Because the structural and dynamical states ($e, \Delta\varphi, R_s, a$) are closed using only dimensionless spectroscopic and chronometric ratios ($e, \theta_\odot, T_{ratio}, z_{sun}$), the variables G and M function as translation factors for legacy unit systems. Their retention is required exclusively for mapping invariant relational geometry onto the kilogram-based convention without increasing the system's predictive capacity.*

Test it yourself

Recommendation: DESMOS
COLAB

7 Time-Density Invariant (τ_D)

Definition 7.1 (Time-Density Invariant)**.** *The time-density invariant τ_D establishes the systemic phase-space limit for any stable trajectory, defined strictly through the orbital period T and visual angular radius θ:*

$$\tau_D = T \sin(\theta)^{\frac{3}{2}} \tag{56}$$

Theorem 7.2 (Relational Density)**.** *The structural density of the central body is determined by the time-density invariant τ_D, bypassing the Newtonian spatial mass distribution.*

Proof. For any stable trajectory, geometry enforces the trigonometric identity $\sin(\theta) = \frac{R_{sur}}{a}$. The operational parameter τ_D algebraically defines the theoretical orbital cycle at the physical surface R_{sur} of the central body:

$$\tau_D = \frac{2\pi R_{sur}}{\beta_{sur} c} \tag{57}$$

where β_{sur} is the local kinematic projection at the surface. Relational density ρ as derived in WILL RG Part I : operates via spatial projection limits:

$$\rho = \frac{\kappa^2 c^2}{8\pi G a^2} \tag{58}$$

Isolating the invariant τ_D structures this relationship as the strict inverse determinant of the system's central density:

$$\tau_D = \sqrt{\frac{\pi}{G\rho}} \tag{59}$$

□

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Theorem 7.3 (Universal Horizon Constant)**.** *At the event horizon boundary limit, the proportional ratio of the limiting time-density $\tau_{D(limit)}$ to the absolute system scale R_s yields a universal horizon constant, operating exclusively on the kinematic limit c and geometric constant π.*

Proof. The systemic absolute scale R_s isolates mathematically from the time-density invariant and surface kinematic projection β_S:

$$R_s = \frac{c}{\pi}\tau_D\beta_S^3 \tag{60}$$

At the event horizon boundary limit ($R_{surf} = R_s$), the potential projection reaches maximum ($\kappa^2 = 1$) and the kinematic projection fixes via the closure theorem (2) ($\beta_S^2 = \frac{1}{2}$). The time-density limit evaluates to:

$$\tau_{D(limit)} = \frac{2\sqrt{2}\pi R_s}{c} \tag{61}$$

The proportional ratio resolves to the universal geometric constant:

$$\boxed{\frac{\tau_{D(limit)}}{R_s} = \frac{2\sqrt{2}\pi}{c}} \tag{62}$$

□

Remark 7.4 (Ontological Consequence)**.** *The algebraic isolation of $\frac{\tau_{D(limit)}}{R_s}$ demonstrates that the boundary mechanics of a spacetime well operate exclusively on geometric proportionality and kinematic cycles. The parameters G and M are completely eradicated from the limiting topological structure of the system.*

8 Algebraic Determination of Absolute System Scale

A classical critique posits that because R_s can be equated to $\frac{2GM}{c^2}$, this derivation is a superficial algebraic reparameterization of Newtonian mechanics. This objection reverses the causal and ontological hierarchy.

The terms G and M are epistemologically secondary composites. They function as empirical translation factors required to map invariant relational geometry onto an assumed absolute spatial container. R_s, conversely, is a primary, coordinate-independent geometric invariant determining the absolute spatial scale of the system's potential projection.

The derivation isolates true dynamic proportionality using exclusively operational observables (chronometric phase T) and intrinsic geometric scaling (a, R_s). Relational Geometry does not inherit from Newtonian constructs; rather, classical mechanics collapses from Relational Geometry when an absolute spatial background is artificially imposed. R_s does not conceal mass; rather, classical mass is an anthropocentric proxy for the invariant geometric boundary R_s.

8.1 Method A: Optical Phase Projection (Angular Radius)

The coordinate of radial distance is systematically eliminated by correlating the local topological phase with the optical geometry of the central body. Any observer extracts the absolute structural parameters of the central body directly from local chronometry and trigonometric projection.

$$\kappa_R^2 = 1 - \frac{1}{(1 + z_{\kappa R})^2}$$

Optical-Topological Invariant (relation between visual geometry and gravitational depth): $\kappa_o^2 = \kappa_R^2 \cdot \sin(\theta_{obs})$

Holographic Chronometry Invariant (surface grazing orbit period): $I_R = T_{obs} \cdot \sin^{\frac{3}{2}}(\theta_{obs})$

Where: θ_{obs} = visual angular radius of the central body observed from the orbital coordinate. κ_R = potential projection at the surface of the central body. I_R = fundamental structural period of the central body (e.g., 10016.33 s for the Solar system). T_{obs} = orbital period of the observer.

The spatial boundaries for the central source surface (r_{surf}) and the observer's orbital semi-major axis (a) are projections of the system scale R_s:

$$r_{surf} = \frac{R_s}{\kappa_R^2}, \quad a = \frac{R_s}{\kappa^2}$$

The exact geometric relationship for the apparent angular radius θ of the central body observed from the orbit nullifies the scale parameter R_s, resulting in a pure ratio of dimensionless projection states:

$$\boxed{\sin(\theta_{obs}) = \frac{\kappa^2}{\kappa_R^2}}$$

This algebraically closes the system, allowing the derivation of the system scale R_s exclusively from local observables (T, θ, z_{sun}) without Newtonian mass approximations:

$$R_s = \frac{Tc}{\pi}(\kappa_R^2 \frac{\sin(\theta_{obs})}{2})^{\frac{3}{2}}$$

And kinetic global form

$$R_s = \frac{Tc}{\pi}\beta^3$$

We now derive algebraic formulas for the system scale R_s using three distinct observational methods. These derivations rely strictly on the **Conservation of the Energy Invariant** W, replacing the need for Newtonian force laws.

8.2 Method B: Differential (Two-Point Method)

This method is suitable when the orbital period is unknown, but the trajectory can be traced geometrically.

Theorem 8.1 (Two-Point Schwarzschild Scale). *Given measurements of geometric position (r) and kinematic intensity (β) at two arbitrary points along a trajectory:*

- *Radii: r_1, r_2 (from astrometry)*
- *Intensities: β_1, β_2 (from de-projected spectral line widths or proper motion)*

the Schwarzschild radius is:

$$\boxed{R_s = \frac{r_1 r_2}{r_2 - r_1}(\beta_1^2 - \beta_2^2).} \tag{63}$$

Proof. We invoke the **Conservation of the Relational Invariant** $W = \frac{1}{2}(\kappa^2 - \beta^2)$. This law states that for any closed system, the specific energy difference between the potential and kinematic projections is constant throughout the orbit.

Therefore, at any two points 1 and 2:

$$\frac{1}{2}(\kappa_1^2 - \beta_1^2) = \frac{1}{2}(\kappa_2^2 - \beta_2^2). \tag{64}$$

Rearranging to group the projections:

$$\kappa_1^2 - \kappa_2^2 = \beta_1^2 - \beta_2^2. \tag{65}$$

Substituting the field identity $\kappa^2 = R_s/r$:

$$\frac{R_s}{r_1} - \frac{R_s}{r_2} = \beta_1^2 - \beta_2^2. \tag{66}$$

Factoring out the scale parameter R_s:

$$R_s(\frac{r_2 - r_1}{r_1 r_2}) = \beta_1^2 - \beta_2^2. \tag{67}$$

Solving for R_s:

$$R_s = \frac{\beta_1^2 - \beta_2^2}{\frac{r_2 - r_1}{r_1 r_2}} = \frac{r_1 r_2}{r_2 - r_1}(\beta_1^2 - \beta_2^2). \tag{68}$$

This formula extracts the "mass" scale purely from geometric gradients. □

8.3 Method C: Geometric Resonance (Balance Point Method)

Using one of two orbital balance points:

- Ascending: $B_a = \arccos(-e)$,
- Descending $B_d = 2\pi - \arccos(-e)$,

when the system passes through its geometric balance points the instantaneous radius r equals the semi-major axis a. At this unique phases, the closure condition $\kappa_o^2 = 2\beta_o^2$ is satisfied.

Theorem 8.2 (Balance Point Formula). *Given the geometric scale a and the total light signal τ measured at the balance point ($r = a$):*

$$\boxed{R_s = \frac{a}{2}(3 - \sqrt{1 + 8\tau^2(B_a)}).} \tag{69}$$

$B_a = \arccos(1-\delta^{-1}) = \arccos(-e) = \arccos(1-\frac{2\beta_p^2}{\kappa_p^2}) = \arccos(\frac{2\beta_a^2}{\kappa_a^2}-1)$ (orbital balance point where $\kappa_o^2 = 2\beta_o^2$ is true)

Proof. **Step 1: Spacetime Factor Expansion.** At the balance point ($r = a$), the closure condition implies $\kappa^2 = R_s/a$ and $\beta^2 = R_s/2a$. The observable τ is:

$$\tau^2 = (1-\kappa^2)(1-\beta^2) = (1-\frac{R_s}{a})(1-\frac{R_s}{2a}). \tag{70}$$

Step 2: Exact Geometric Solution. Expanding the product:

$$\tau^2 = 1 - \frac{R_s}{2a} - \frac{R_s}{a} + \frac{R_s^2}{2a^2} = 1 - \frac{3R_s}{2a} + \frac{R_s^2}{2a^2}. \tag{71}$$

Multiplying by $2a^2$ to clear denominators and rearranging into standard quadratic form ($Ax^2 + Bx + C = 0$):

$$R_s^2 - 3aR_s + 2a^2(1-\tau^2) = 0. \tag{72}$$

Solving for R_s using the quadratic formula ($x = \frac{-b\pm\sqrt{b^2-4ac}}{2a_{coef}}$):

$$R_s = \frac{3a \pm \sqrt{(3a)^2 - 4(1)(2a^2(1-\tau^2))}}{2}. \tag{73}$$

Simplifying the term under the radical:

$$9a^2 - 8a^2(1-\tau^2) = 9a^2 - 8a^2 + 8a^2\tau^2 = a^2(1+8\tau^2). \tag{74}$$

Thus:

$$R_s = \frac{3a \pm a\sqrt{1+8\tau^2}}{2} = \frac{a}{2}(3 \pm \sqrt{1+8\tau^2}). \tag{75}$$

For stable orbits, we must select the negative root. □

Remark 8.3 (Light Separation). *This unit-less factor in brackets is:*

- ***Potential projection at semi-major axis:*** $\kappa^2 = \frac{1}{2}(3 - \sqrt{1+8\tau^2(B_a)})$

A unique way to separate gravitational part from the light signal.

8.4 Method D: Instantaneous (Arbitrary Phase Method)

Most generally, if the orbital geometry (a) is known, the scale R_s can be derived from a **single epoch** observation at any arbitrary radius r_o.

Theorem 8.4 (Arbitrary Phase Formula). *Given the orbital geometry (a, r_o) and the single light observable τ_{W_o}:*

$$\boxed{R_s = \frac{r_o}{2(2a-r_o)}(4a - r_o - \sqrt{(4a-r_o)^2 - 8a(2a-r_o)(1-\tau_o^2)}).} \tag{76}$$

Proof. **Step 1: Relational Invariant Form.** We start with the energy invariant relation $W = \frac{1}{2}(\kappa^2 - \beta^2)$. For a bound orbit, $W = R_s/4a$. We express the kinematic intensity β^2 at arbitrary radius r in terms of the field intensity κ^2:

$$\frac{1}{2}(\kappa^2-\beta^2) = \frac{R_s}{4a} \quad \Longrightarrow \quad \beta^2 = \kappa^2 - \frac{R_s}{2a}. \tag{77}$$

Substituting $\kappa^2 = R_s/r$:

$$\beta^2 = R_s(\frac{1}{r} - \frac{1}{2a}). \tag{78}$$

Step 2: Constraint via Observables. We substitute the expressions for κ^2 and β^2 into the exact observable constraint $\tau_o^2 = (1-\kappa^2)(1-\beta^2)$:

$$\tau_o^2 = (1-\frac{R_s}{r})(1-R_s[\frac{1}{r}-\frac{1}{2a}]). \tag{79}$$

Let $A = \frac{1}{r}$ and $B = \frac{1}{r} - \frac{1}{2a}$. The equation becomes:

$$\tau_o^2 = (1-AR_s)(1-BR_s) = 1-(A+B)R_s + ABR_s^2. \tag{80}$$

Rearranging into quadratic form:

$$(AB)R_s^2 - (A+B)R_s + (1-\tau_o^2) = 0. \tag{81}$$

Step 3: Coefficient Expansion. We compute the coefficients explicitly:

$$AB = \frac{1}{r}(\frac{2a-r}{2ar}) = \frac{2a-r}{2ar^2} \tag{82}$$

$$A+B = \frac{1}{r} + \frac{2a-r}{2ar} = \frac{2a+2a-r}{2ar} = \frac{4a-r}{2ar} \tag{83}$$

Multiplying the entire quadratic equation by $2ar^2$ to clear denominators:

$$(2a-r)R_s^2 - r(4a-r)R_s + 2ar^2(1-\tau_o^2) = 0. \tag{84}$$

Step 4: Solution. Solving for R_s:

$$R_s = \frac{r(4a-r) \pm \sqrt{r^2(4a-r)^2 - 4(2a-r)(2ar^2)(1-\tau_o^2)}}{2(2a-r)}. \tag{85}$$

Factoring out r^2 from the radical term $\sqrt{r^2(...)} = r\sqrt{(...)}$:

$$R_s = \frac{r}{2(2a-r)}((4a-r) \pm \sqrt{(4a-r)^2 - 8a(2a-r)(1-\tau_o^2)}). \tag{86}$$

For stable orbits, we select the negative root, yielding the exact algebraic link between the observed light phase and the system's geometric scale. □

9 Rotational Systems (Kerr Without Metric)

Contextual Bounds

- **For a gravitationally closed (static) system,** the physical boundary is defined by the condition $\kappa^2 = 1$. The closure principle ($\kappa^2 = 2\beta^2$) is what dictates that this corresponds to a kinetic state of $\beta^2 = 1/2$.
- **For a kinematically closed (maximally rotating) system,** the physical boundary is defined by the condition $\beta^2 = 1$. The same closure principle ($\kappa^2 = 2\beta^2$) then necessitates that the corresponding gravitational state must be $\kappa^2 = 2$.

For rotating black holes, we establish the connection between relational kinetic projection and the Kerr metric by defining:

$$\beta = \frac{ac^2}{GM}, \quad \kappa = \sqrt{2}\beta$$

where:

- β is the relational rotation parameter, with $0 \leq \beta \leq 1$,
- κ is related to the geometry and gravity,
- $R_s = \frac{2GM}{c^2}$ is the Schwarzschild radius,
- $a = \frac{J}{Mc}$ is the Kerr rotation parameter,
- J is the angular momentum of the black hole,
- M is the mass of the black hole.

We also derive a key invariant relationship:

$$a_{\max} = \frac{R_s}{2} = \beta_{\max}^2 r$$

This relationship holds when $r = \frac{R_s}{2\beta^2}$, providing an elegant connection between the parameters.

Event Horizon

Using our approach, the inner and outer event horizons of the Kerr metric are expressed as:

$$r_{\pm} = \frac{R_s}{2}(1 \pm \beta_Y)$$

For the extreme case where $\beta = 1$ (maximal rotation), the horizons merge at:

$$r_+ = r_- = \frac{R_s}{2}$$

This coincides with the minimum radius in our model predicted using maximum value of κ parameter $\kappa_{max} = \sqrt{2}$:

$$r_{\min} = \frac{1}{\kappa_{max}^2} R_s = \frac{1}{2} R_s$$

Ergosphere

The radius of the ergosphere in our model is described as:

$$r_{\text{ergo}} = \frac{R_s}{2}(1 + \sqrt{1 - \beta^2 \cos^2 \theta})$$

This formulation correctly reproduces the key features of the ergosphere:

- At the equator ($\theta = \pi/2$), $r_{\text{ergo}} = R_s$ for any rotation parameter,
- At the poles ($\theta = 0$), r_{ergo} coincides with the event horizon radius.

Ring Singularity

Unlike the Schwarzschild metric with its point singularity, the Kerr metric features a ring singularity located at:

$$r = 0, \quad \theta = \frac{\pi}{2}$$

The "size" of this ring is proportional to $a = \frac{GM}{c^2}\beta$, reaching its maximum for extreme black holes ($\beta = 1$).

Naked Singularity

For $\beta \leq 1$, a naked singularity does not emerge, aligning with the cosmic censorship Principle. In our model, Energy Symmetry Law enforce constraint by limiting β to the range $[0, 1]$.

9.1 Chiral Asymmetry: The Algebraic Origin of Frame-Dragging

In standard General Relativity (GR), the chiral asymmetry of orbits around a rotating mass—commonly termed "frame-dragging" or the Lense-Thirring effect—is structurally derived from the off-diagonal $g_{t\phi}$ component of the Kerr metric tensor. In Relational Orbital Mechanics (R.O.M.), this asymmetry is not a geometric deformation of a background manifold, but a strict algebraic consequence of linear phase superposition on the directed S^1 carrier.

Composite Kinematic Projection and DOF Conservation

Because the kinematic projection β operates on the 1-DOF relational carrier S^1, motions sharing the same directional axis must superimpose linearly prior to quadratic energy evaluation. For a test mass orbiting a spinning central body, the composite kinematic state is:

$$\beta_\Sigma = \beta_{\text{orb}} \pm \beta_{\text{spin}} \tag{87}$$

where $(+)$ denotes prograde and $(-)$ denotes retrograde motion.

To conserve dimensional degrees of freedom when projecting a 3D central body's spin onto the 1D carrier S^1, a 1/3 scaling factor is applied. The local operational spin projection scales the specific angular momentum $a = J/(Mc)$ against the spatial depth r:

$$\beta_{\text{spin}} = \frac{a}{3r} \tag{88}$$

The Algebraic Symmetry Breaker

Applying the relational closure theorem ($\kappa^2 = 2\beta^2$) strictly to the composite state yields:

$$\kappa^2 = 2(\beta_{\text{orb}} \pm \beta_{\text{spin}})^2 = 2\beta_{\text{orb}}^2 + 2\beta_{\text{spin}}^2 \pm 4\beta_{\text{orb}}\beta_{\text{spin}} \tag{89}$$

The emergent cross-term $\pm 4\beta_{\text{orb}}\beta_{\text{spin}}$ is the exact symmetry breaker. Because $\kappa^2 = R_s/r$, this mathematically mandates that a prograde orbit $(+)$ requires a deeper potential projection (smaller radius) than a retrograde orbit $(-)$ at an equivalent orbital velocity. This algebraic constraint is operationally indistinguishable from the spatial drag generated by the $g_{t\phi}$ metric.

Expansion of the Precessional Divergence

The internal causal order of the composite system evaluates via the relational phase factor $\tau^2 = (1-\kappa^2)(1-\beta_\Sigma^2)$. Expanding this algebraically and applying the strict directional boundary ($\pm = 1$) isolates the chiral divergence:

$$\tau^2_{Y(\text{chiral})} = \pm(6\beta_{\text{orb}}\beta_{\text{spin}} - 8\beta_{\text{orb}}^3\beta_{\text{spin}} - 8\beta_{\text{orb}}\beta_{\text{spin}}^3) \tag{90}$$

Substituting the local operational projection into the isolated chiral polynomial defines the phase shift per orbital cycle, normalized by the orbital shape $(1-e^2)$:

$$\Delta\varphi_{\text{chiral}} = \frac{2\pi}{1-e^2}\left[\pm(6\beta_{\text{orb}}\beta_{\text{spin}} - 8\beta_{\text{orb}}^3\beta_{\text{spin}} - 8\beta_{\text{orb}}\beta_{\text{spin}}^3)\right] \tag{91}$$

9.2 Legacy Translation: Lense-Thirring Equivalence

To verify structural compatibility with historical models, we isolate the primary geometric observable by ablating the higher-order geometric cross-coupling terms ($\beta^4 \to 0$) and setting the orbit to circular ($e = 0$). The equation simplifies algebraically to:

$$\Delta\varphi_{\text{ablated}} \simeq 12\pi\beta_{\text{orb}}\beta_{\text{spin}} \tag{92}$$

Substituting the operational spin projection $\beta_{\text{spin}} = \frac{J}{3Mcr}$ and the kinematic orbital projection $\beta_{\text{orb}} = \frac{v}{c}$ recovers the empirical post-Newtonian node precession:

$$\Delta\varphi_{\text{ablated}} \simeq 12\pi\left(\frac{v}{c}\right)\left(\frac{J}{3Mcr}\right) = \frac{4\pi vJ}{Mc^2r} \tag{93}$$

Applying the kinematic scaling identity $\frac{v}{M} = \frac{G}{rv}$:

$$\Delta\varphi_{\text{ablated}} \simeq \frac{4\pi GJ}{c^2r^2v} \tag{94}$$

The Lense-Thirring effect emerges natively from the algebraic closure of the relational carriers, executed without invoking differential tensors or a curved pseudo-Riemannian manifold.

COLAB EMPIRICAL VERIFICATION

This Chiral Asymmetry when tested against LAGEOS-1 Satellite (Earth orbit) shows perfect agreement with observations and GR predictions: LAGEOS I Lense-Thirring.ipynb

9.3 Epistemological Conclusion

Mathematics dictates that if two formalisms enforce identical symmetries - conservation of the relational energy budget and causal limit - they must converge on identical structural polynomials. R.O.M. achieves this convergence without postulating differential tensors or a curved pseudo-Riemannian manifold. Frame-dragging is thus revealed not as a mechanical torsion of spacetime, but as the algebraic consequence of squaring a co-linear phase superposition under the relational closure bound $\kappa^2 = 2\beta^2$.

Physical Interpretation

- **No need for pre-existing spacetime** - geometry emerges from energy distributions.
- **All parameters** are dimensionless and directly derived from the speed of light as finite resource.
- **Scale invariance:** The same structure applies from Planck-scale objects to galactic black holes.

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10 Relational Derivation of the L1 Equilibrium Point

The calculation of the Sun–Earth L1 equilibrium point is performed strictly through relational parameters derived from direct observables, independent of mass M or the gravitational constant G.

10.1 Observational Primitives

The derivation utilises only the following epistemologically direct quantities (identical in form to those used for Mercury and S2 in earlier sections):

- $\theta_\odot$: angular radius of the Sun (0.0046524 rad).
- $z_\odot$: spectroscopic fractional frequency shift at the solar surface (2.1224×10^{-6}).
- T_E: sidereal orbital period of the Earth (3.15576×10^{7} s).
- T_m: sidereal orbital period of the Moon (2.3606×10^{6} s).
- R_m: mean Earth–Moon distance (3.844×10^{8} m).

10.2 Structural Parameters from Relational Projections

Using the absolute-scale formulae established in Section 8 (Method A) and the global kinematic relation $R_s = Tc\beta^3/\pi$:

$$R_{s\odot} = \frac{T_E}{\pi} c \left(\sqrt{\frac{1}{2} \left(1 - \frac{1}{(1+z_\odot)^2} \right) \sin(\theta_\odot)} \right)^3, \tag{95}$$

$$\beta_m = \frac{2\pi R_m}{T_m c}, \tag{96}$$

$$R_{s\mathrm{E}} = \frac{T_m}{\pi} c\, \beta_m^3, \tag{97}$$

$$R_E = \frac{R_{s\odot}}{\left(1 - \frac{1}{(1+z_\odot)^2}\right) \sin(\theta_\odot)}. \tag{98}$$

These expressions recover the Earth–Sun distance $R_E \approx 1.496 \times 10^{11}$ m and the Earth/Sun scale ratio $\mu = R_{s\mathrm{E}}/R_{s\odot} \approx 3.003 \times 10^{-6}$ directly from the observables, without any reference to Newtonian mass.

10.3 Earth–L1 Distance Derivation

Let R_{L1} be the distance from Earth to the L1 point and define the normalised separation

$$\alpha = \frac{R_E - R_{L1}}{R_E}.$$

Synchronous motion at L1 requires the kinematic projection β_{L1} to satisfy the orbital-period constraint. Enforcing the closure theorem $\kappa^2 = 2\beta^2$ together with energy symmetry yields the exact quintic

$$\alpha^5 - 2\alpha^4 + \alpha^3 + \left(\frac{R_{s\mathrm{E}}}{R_{s\odot}} - 1\right)\alpha^2 + 2\alpha - 1 = 0.$$

In the limit $R_{s\mathrm{E}} \ll R_{s\odot}$ the physical root is

$$R_{L1} \approx R_E \sqrt[3]{\frac{R_{s\mathrm{E}}}{3R_{s\odot}}}.$$

Substituting the numerical values obtained above gives

$$R_{L1} \approx 1.498 \times 10^{9}\ \mathrm{m}$$

(with more precise ephemeris constants the value rises to 1.501×10^{9} m). Both figures lie within the combined uncertainty of the input observables $z_\odot$ and $\theta_\odot$.

Result

The relational derivation reproduces the observed Earth–L1 distance

$$R_{L1} \approx 1.5 \times 10^9\,\mathrm{m}$$

to better than 0.3% without invoking G, M, or a background metric. The same algebraic chain that recovers Mercury's precession and the S2 orbit now also locates the collinear equilibrium point of the Sun–Earth–test-mass system.

Test it yourself

Full symbolic + numerical verification (including the exact quintic root): L1 Relational Derivation.ipynb

This section closes the demonstration that every major gravitational observable — two-body orbits, precession, light deflection, and now the three-body L1 point — emerges as an inevitable algebraic consequence of the Foundational Methodological Principles. All WILL RG results including the complete success of R.O.M. remains a single unbroken chain of derivations from firs principles under extreme methodological constrains and zero flexibility.

11 Discussion

Relational Orbital Mechanics is not an independent model. It is the direct operational consequence of applying the Foundational Methodological Principles and the Unifying Ontological Principle of WILL Relational Geometry to bound gravitational systems. The entire content of this work - from the algebraic classification of allowed orbital states to the derivation of the Sun–Earth L1 contour - follows from a single generative chain that begins with the refusal to introduce unjustified background structures and continues with the closure condition $\kappa^2 = 2\beta^2$ as a necessary geometric identity rather than an empirical input.

11.1 Epistemic and Operational Benefits

Operational Benefits: Absence of Differential Formalism and Acceleration

The most immediate operational consequence of the relational construction is that **no differential equations of motion and no concept of acceleration are required**. In standard orbital mechanics the central acceleration (Newtonian or geodesic) is the primitive from which orbital period, shape, precession, and equilibrium configurations are derived. In WILL RG the same observables are obtained directly from algebraic identities among the projections on the carriers S^1 and S^2, once the methodological principles and the unifying relation SPACETIME $\equiv$ ENERGY have fixed the geometry of those carriers and the unique exchange rate between them.

This absence is not a simplification imposed after the fact; it follows necessarily from the generative structure. Because the theory begins by removing the separation between structure and dynamics, and because it enforces topological closure and maximal symmetry from the outset, the allowed relational states of a bound system are completely classified by the dimensionless projections and the single closure condition that the methodology itself produces. No integration of trajectories, no effective potentials, and no auxiliary constructs such as rotating frames are needed. The mathematics remains a short sequence of explicit algebraic relations whose every step retains clear physical meaning.

The gain in transparency is therefore structural. Observables that in conventional treatments appear only after lengthy perturbative expansions or numerical integration of differential equations emerge here as immediate, exact consequences of the relational geometry. This is visible across all derivations in the paper, including the period–semi-major-axis relation, eccentricity, secular precession, light deflection, and the location of the Sun–Earth L1 point (Sections 6 and 10).

11.2 Challenges and Current Limitations

While the algebraic closure of Relational Orbital Mechanics accounts for the complete phenomenology of bound two-body systems and the collinear restricted three-body configuration, two important domains remain outside the present treatment.

Gravitational waves and the full radiative two-body problem have not yet been formulated within the relational framework. The current construction addresses only conservative, bound orbits. Extending it to dissipative, radiative dynamics requires a relational description of propagating degrees of freedom on the carriers and remains an open technical task.

A complete, self-consistent multi-body architecture that extends beyond the collinear L1 case has likewise not been constructed. The nested-contour approach provides a clear conceptual route, but its explicit algebraic realisation for systems containing three or more gravitationally bound bodies (including mean-motion resonances and long-term stability) requires further development. The author states openly that progress on both fronts has so far been limited by present mathematical facility rather than by any intrinsic prohibition within the relational ontology. Collaboration with researchers experienced in radiative dynamics and in the algebraic treatment of multi-body closures is therefore actively invited.

11.3 Falsifiability and Future Tests

The framework yields several sharp, near-term testable predictions that follow directly from its algebraic structure.

The most immediate discriminator is continued high-precision monitoring of the S2 star. The relational model predicts that the true-anomaly-coupled precession parametrisation will preserve or strengthen its statistical advantage over the conventional time-linear implementation as new pericentre passages are added with improved GRAVITY+ astrometry (2026–2028 and subsequent windows). A statistically significant reversal of the present $\Delta\chi^2$ improvement on the enlarged dataset would falsify the relational phasing hypothesis while remaining compatible with the magnitude of the precession itself.

For Solar-System tests, the explicit second-order correction in the R.O.M. precession formula is in principle accessible with sufficiently long observational baselines or with planets at smaller semi-major axes. A confirmed detection at the predicted magnitude, or a firm upper bound below it, would directly constrain the cross-coupling term that is absent from the first-order post-Newtonian expression.

Finally, the relational derivation of the L1 contour predicts a specific, observationally accessible offset between the classical Hill radius and the true equilibrium surface. Precision station-keeping telemetry from future L1 halo missions can test whether the location of minimal ΔV coincides with the predicted relational contour.

These tests are decisive because they probe either the phasing of precession or the explicit higher-order content of the closure condition — both of which are uniquely fixed by the relational algebra and are not adjustable parameters.

12 Conclusion

The Relational Orbital Mechanics system demonstrates that the entire phenomenology of motion – from Keplerian architecture to perihelion precession and light deflection – emerges as an inevitable algebraic consequence of the Foundational Methodological Principles. All WILL RG results including the complete success of R.O.M. remains an unbroken chain of derivations from firs principles under extreme methodological constrains with zero flexibility.

a handful of dimensionless relational projections and the single, topologically forced closure condition $\kappa^2 = 2\beta^2$. No background manifold, metric tensor, or independent constants of nature beyond the speed of light are required.

This is not a re-description of General Relativity; it is the irreducible algebraic core that remains when the ontological superstructure of Riemannian geometry and tensorial formalism is removed. The fact that the S2 orbit is fit *better* by the phase-coupled precession parametrization inherent to R.O.M. – with no additional free parameters – indicates that the relational approach is not merely philosophically appealing but empirically preferred in the strong-field regime.

Within R.O.M. several structural consequences follow:

1. **G and M are secondary.** The Schwarzschild length R_s and all orbital distances are derived directly from dimensionless spectroscopic and chronometric ratios. The Newtonian gravitational constant and the kilogram mass appear only as conversion factors when translating geometric results into legacy units.

2. **Closure replaces the virial theorem.** The relation $\kappa^2 = 2\beta^2$ is not an empirical input but the inevitable consequence of equitable budget distribution across the independent degrees of freedom of the kinematic and potential carriers.

3. **Precession is topological, not dynamical.** The secular advance of the pericentre arises from the mismatch of internal phase over one cyclic extension of the orbital phase, without integrating geodesic equations or expanding in powers of v/c.

4. **The event horizon can be interpret as relational lockout.** When the periapsis precession approaches orbital angular speed the trajectory folds onto itself and ceases to advance in angular phase. This is the geometric origin of the horizon, free of singular coordinates.

WILL trilogy offers a unified, parameter-free ground-up unbroken chain derivation of most major observed phenomena within extreme methodological constrains and 0 flexibility. The traditional unjustified separation of spacetime and energy dissolves into a single generative structure we call WILL.

The results presented here invite exacting numerical tests (additional binary pulsars, triple systems, high-resolution S2 pericentre passages) and provide an operational language in which gravitational phenomena can be communicated without surplas ontology. The hope is that the physical clarity returned by algebraic transparency will help turn the focus of gravitational science back to its earliest ambition: to see the logic of the world, not just to model it.

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